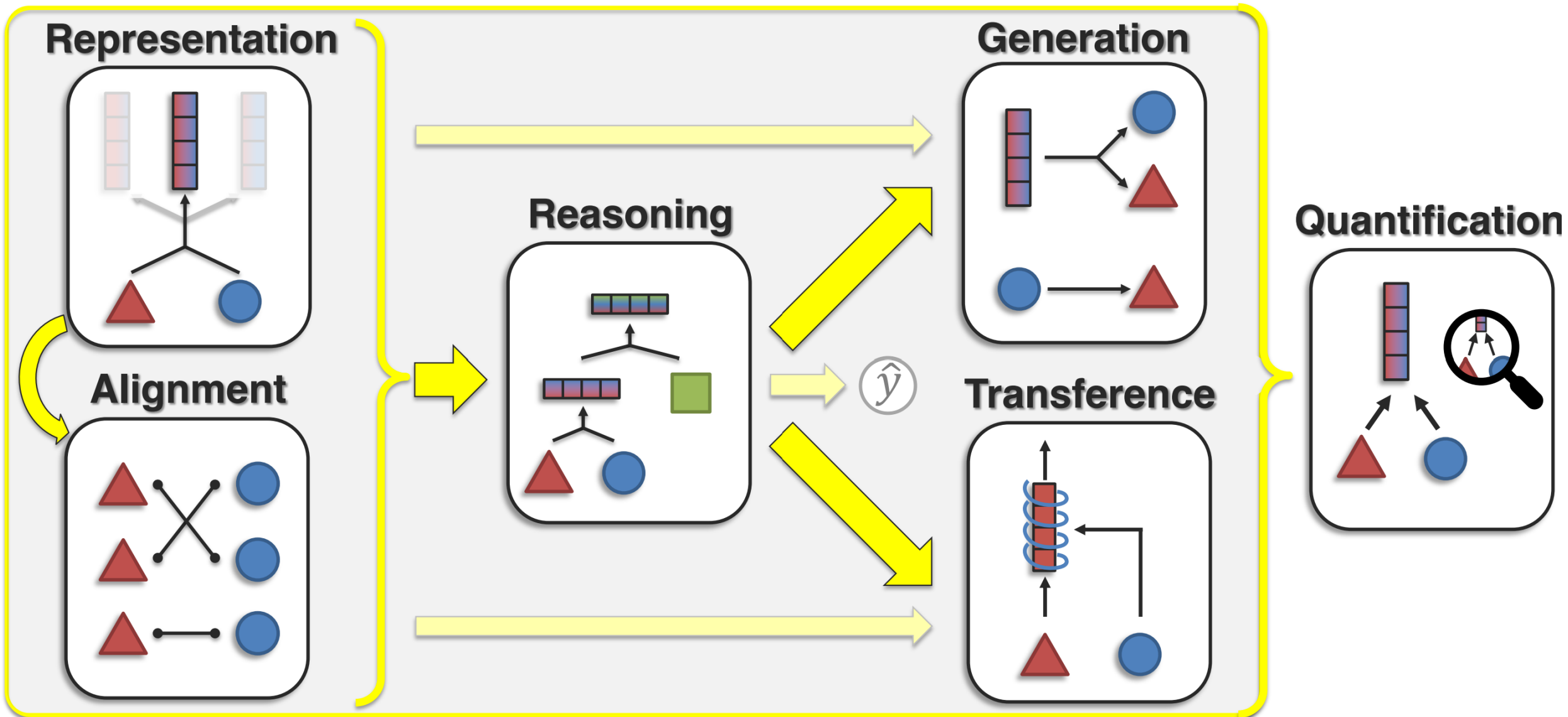


Multimodal Machine Learning

Lecture 11.2: Transference

Dam Quang Tuan

Core Multimodal Challenges

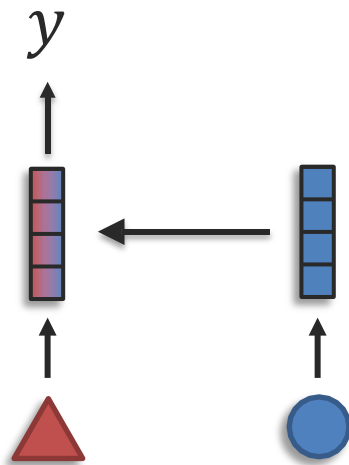


Transference

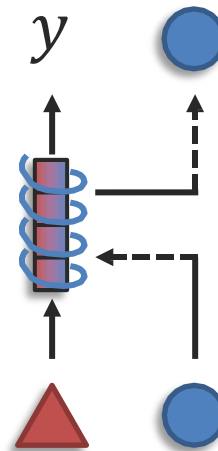
Definition: Transfer knowledge between modalities, usually to help the primary modality which may be noisy or with limited resources

Sub-challenges:

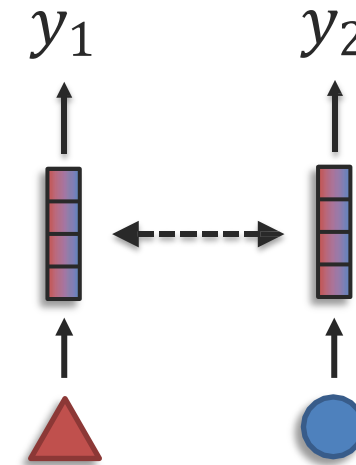
Transfer



Co-learning

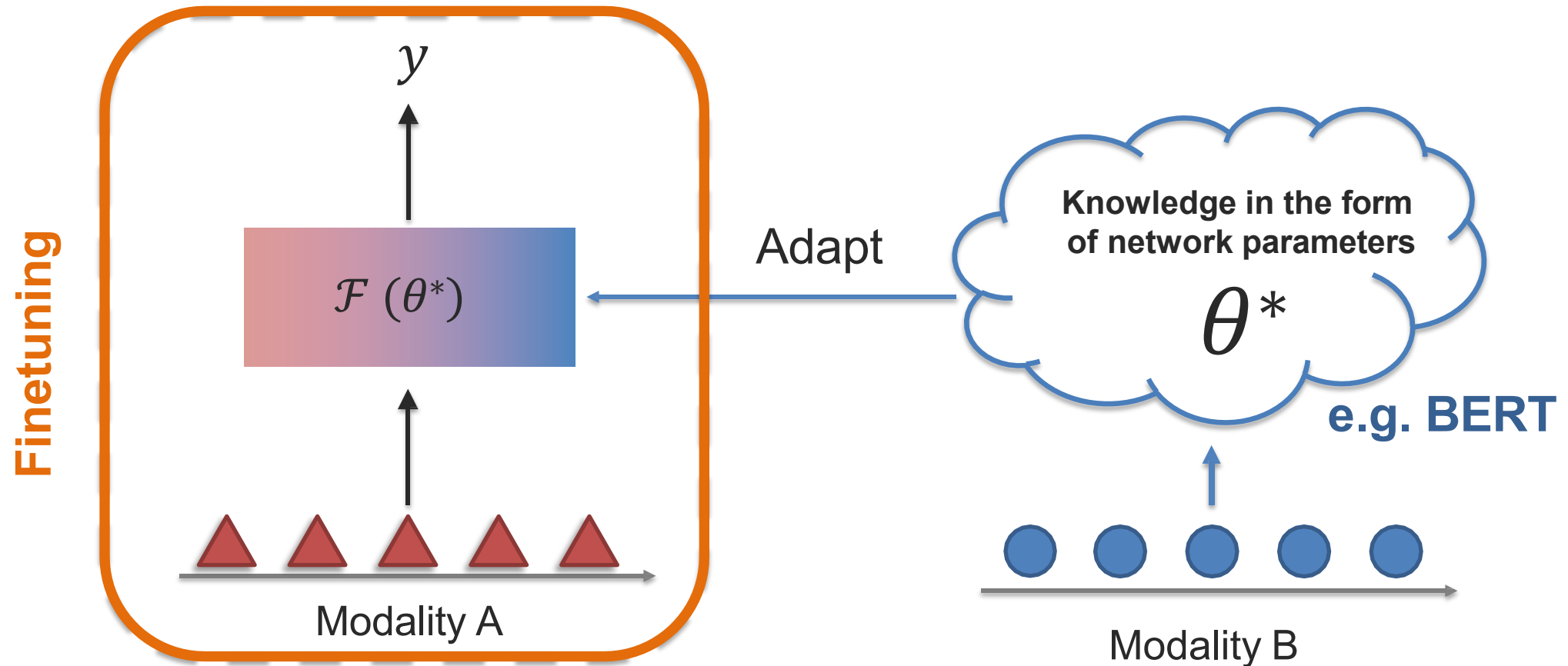


Model Induction



Sub-Challenge 5a: Transfer via Pretrained Models

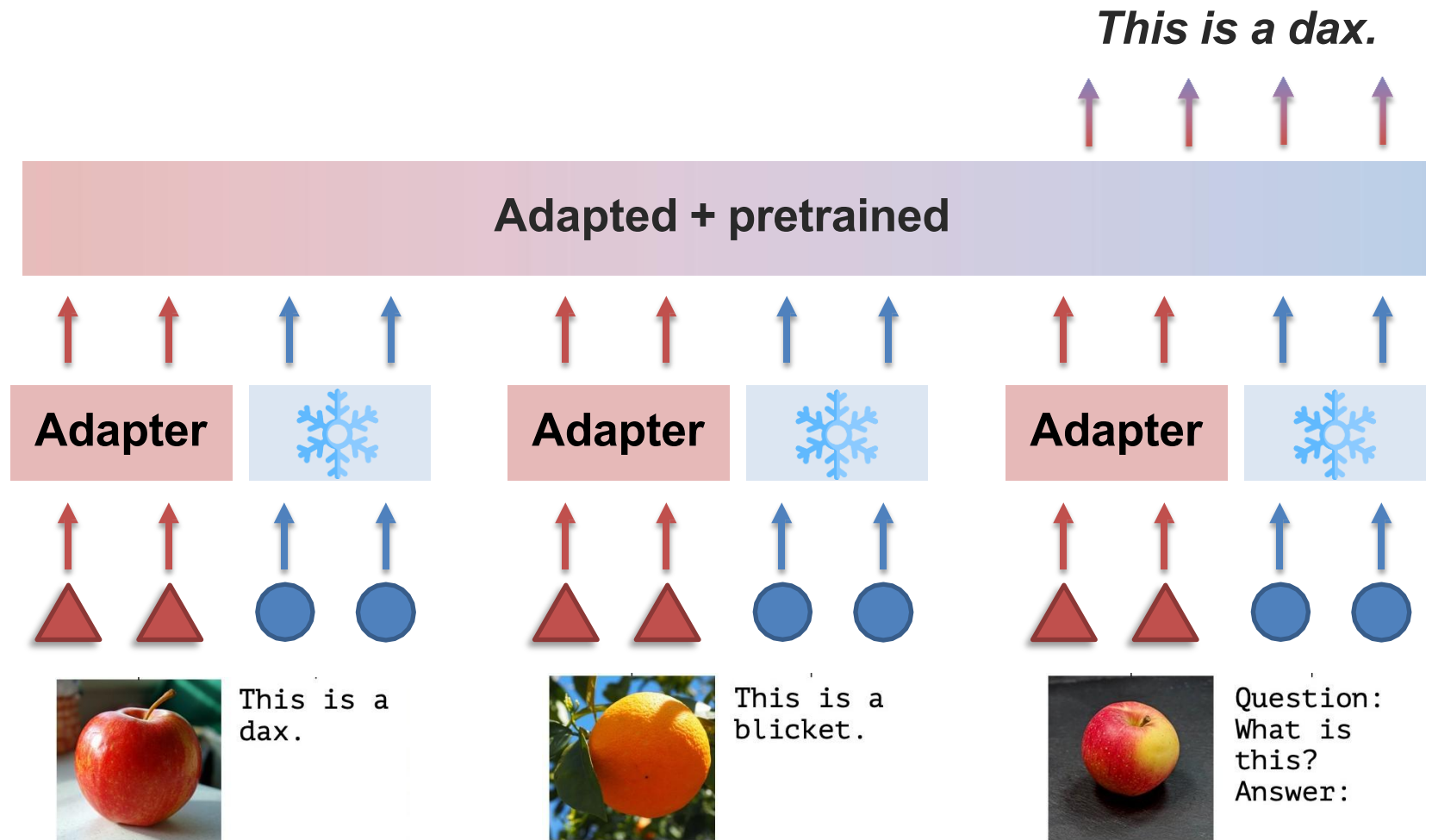
Definition: Transferring knowledge from large-scale pretrained models to downstream tasks involving the primary modality.



Sub-Challenge 5a: Transfer via Pretrained Models

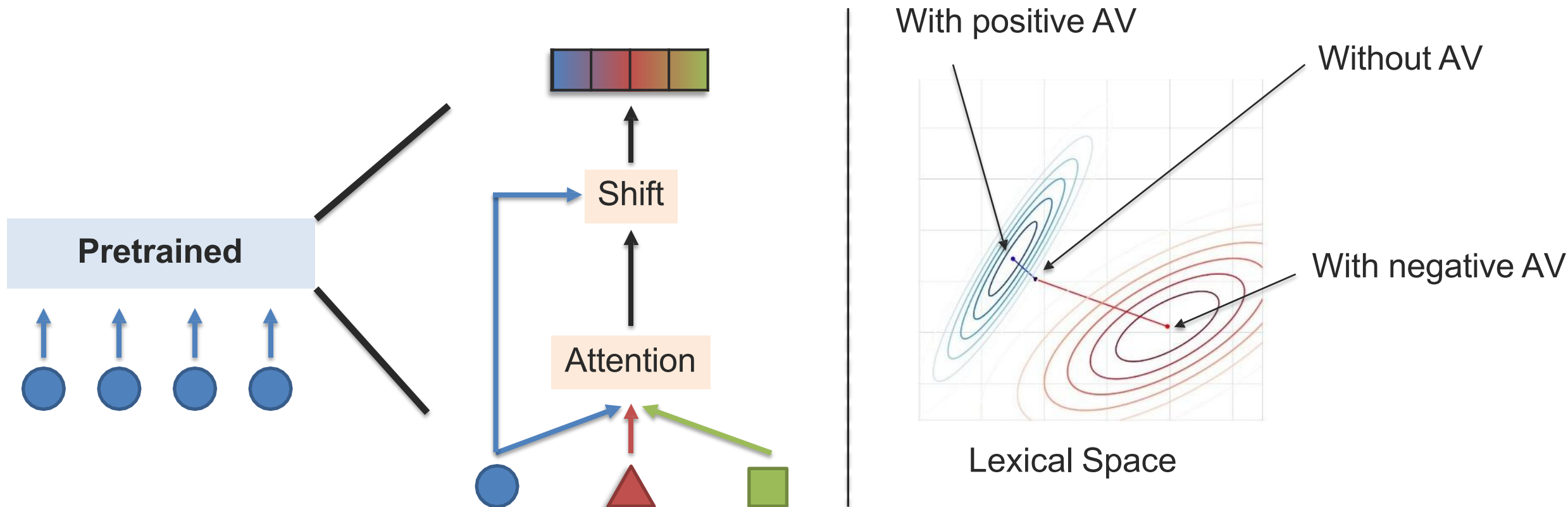
Transfer via prefix tuning

Few-shot image classification:



Sub-Challenge 5a: Transfer via Pretrained Models

Transfer via representation tuning



[Ziegler et al., Encoder-Agnostic Adaptation for Conditional Language Generation. arXiv 2019]

[Rahman et al., Integrating Multimodal Information in Large Pretrained Transformers. ACL 2020]

Sub-Challenge 5a: Transfer via Pretrained Models

1. Disentanglement

$$\mathcal{L}_\beta(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - \beta \cdot \text{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$

2. Conditioning

$$p(\mathbf{x}_{0:T} \mid y) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t, y)$$

3. Prompt tuning

4. Representation tuning

5. Classifier gradient tuning

$$\nabla \log p(\mathbf{x}_t \mid y) = \underbrace{\nabla \log p(\mathbf{x}_t)}_{\text{unconditional score}} + \gamma \underbrace{\nabla \log p(y \mid \mathbf{x}_t)}_{\text{classifier gradient}}$$

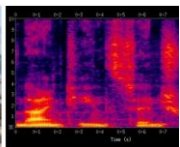
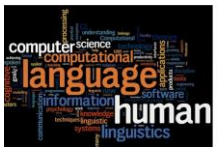
6. Classifier-free tuning

$$\nabla \log p(\mathbf{x}_t \mid y) = \underbrace{\gamma \nabla \log p(\mathbf{x}_t \mid y)}_{\text{conditional score}} + \underbrace{(1 - \gamma) \nabla \log p(\mathbf{x}_t)}_{\text{unconditional score}}$$

Multitask and Transfer Learning

How can we transfer knowledge across multiple tasks, each over a different subset of modalities?

Video
classification

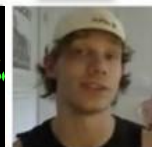


Language

Video

Audio

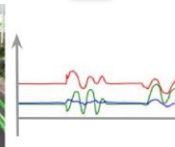
Sentiment,
emotions



Audio

Video

Robot
dynamics



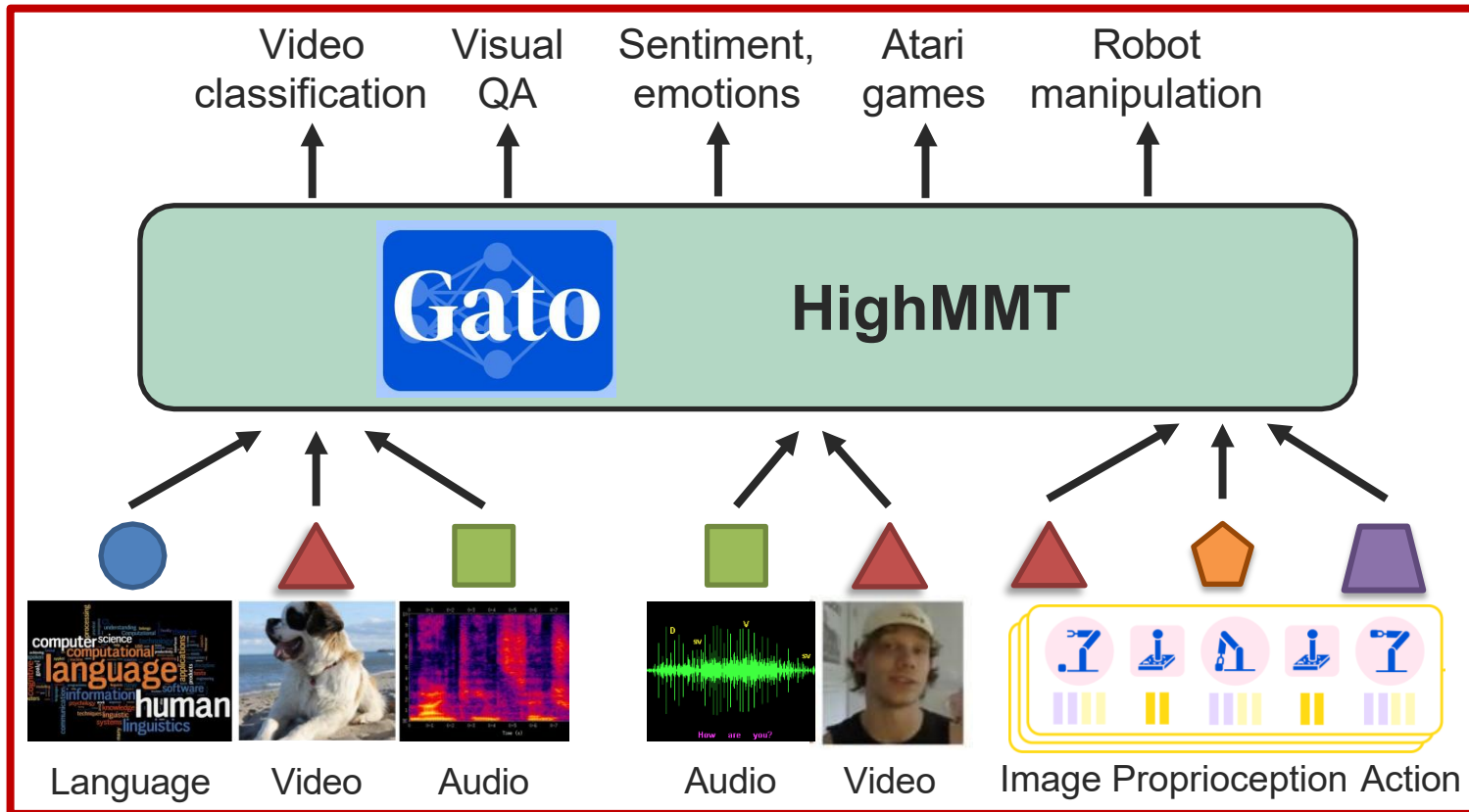
Video Time-series

Generalization across modalities and tasks
Important if some tasks are low-resource

High-Modality Multimodal Transformers

Transfer across partially observable modalities

Unified model + parameter sharing + multitask and transfer learning



Non-parallel multitask learning

Task-specific classifiers

Same model architecture!

Shared multimodal model

Same parameters!

Modality-specific embeddings

Standardized input sequence

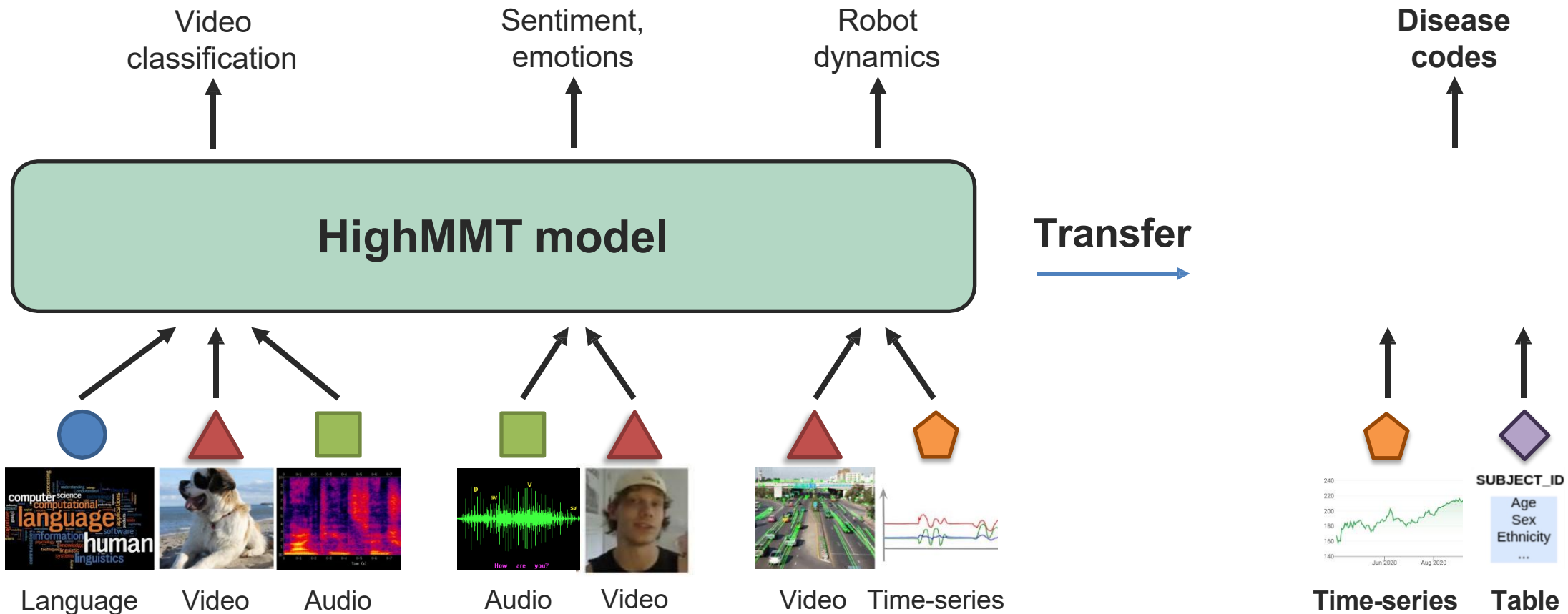
[Reed et al., A Generalist Agent. TMLR 2022]

[Liang et al., HighMMT: Quantifying Modality and Task Heterogeneity for High-Modality Representation Learning. TMLR 2022]

Multitask and Transfer Learning

Transfer across partially observable modalities

HighMMT: unified model + parameter sharing + multitask and transfer learning

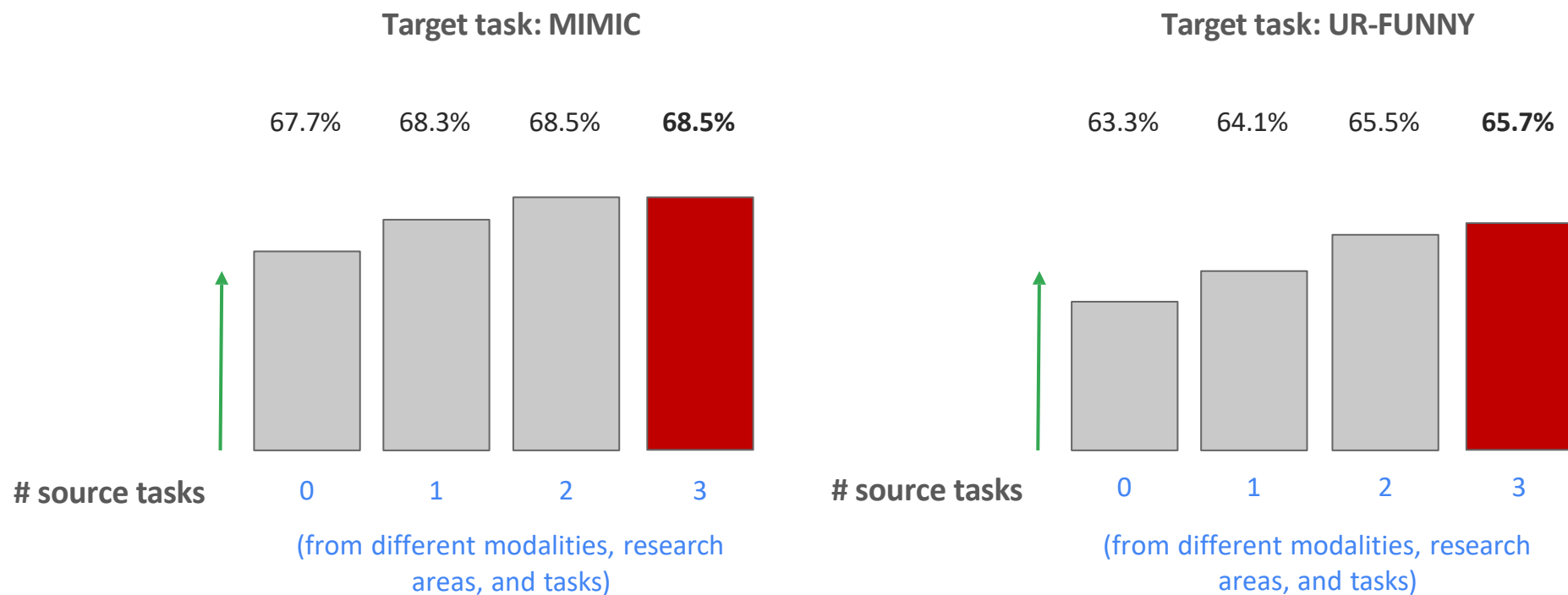


[Liang et al., HighMMT: Towards Modality and Task Generalization for High-Modality Representation Learning. TMLR 2022]

Multitask and Transfer Learning

Transfer across partially observable modalities

HighMMT: unified model + parameter sharing + multitask and transfer learning



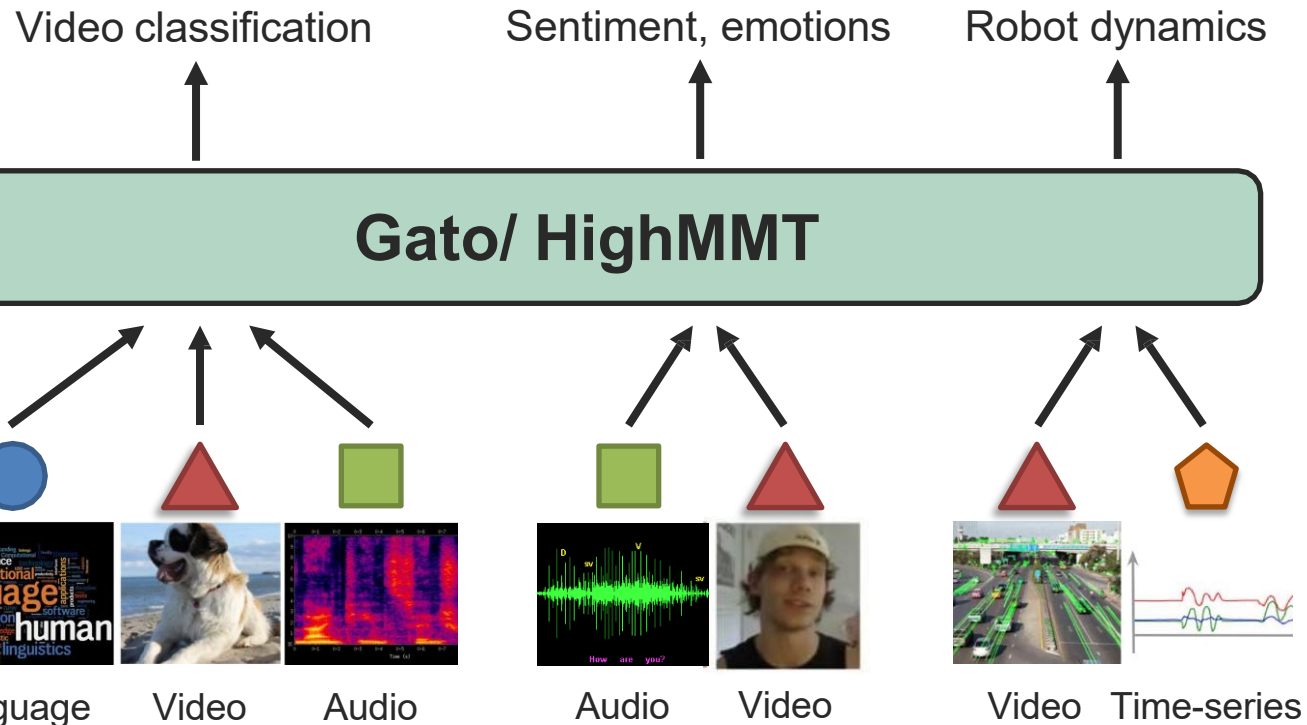
Achieves both multitask and transfer capabilities across modalities and tasks

High-Modality Models

Open
challenges

Some implicit assumptions:

- All modalities can be represented as sequences without losing information.
- Dimensions of heterogeneity can be perfectly captured by modality-specific embeddings.
- Cross-modal connections & interactions are shared across modalities and tasks.



Shared multimodal model?

Modality-specific embeddings?

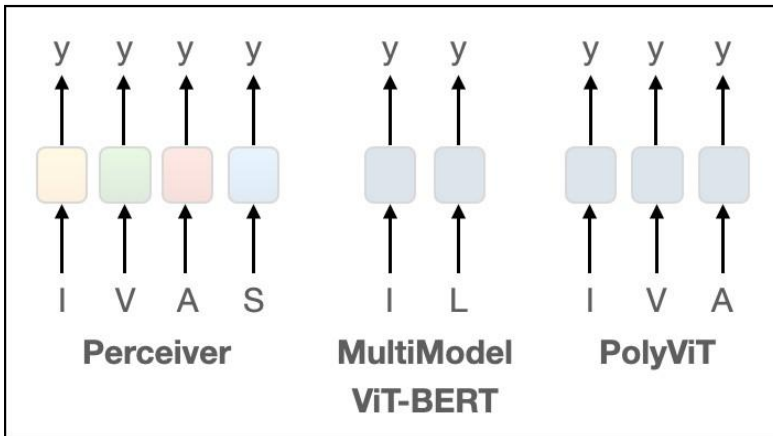
Standardized input sequence?

Multitask and Transfer Learning

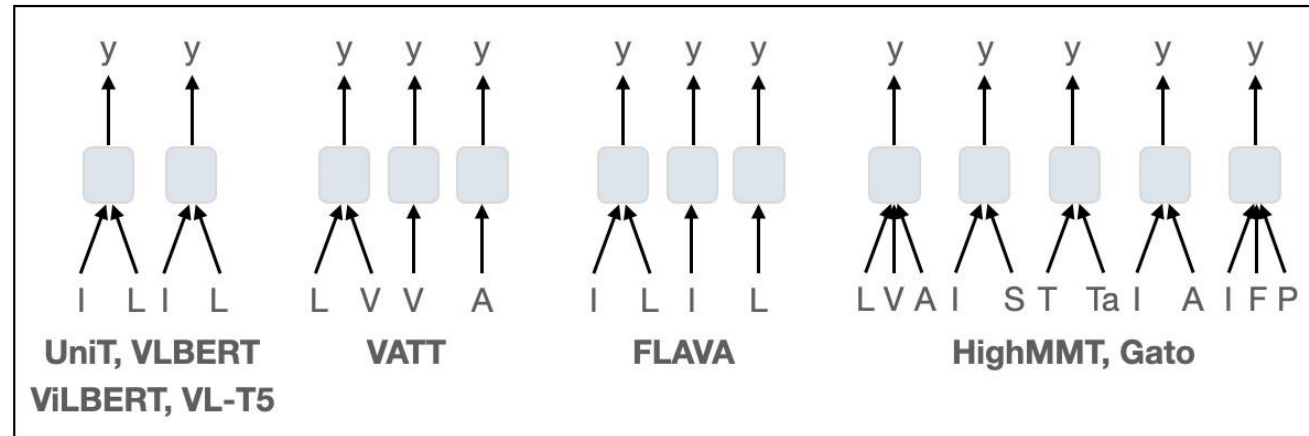
Open
challenges

Many more dimensions of transfer

Unified encoder for unimodal learning



Multimodal multitask learning



I: image
V: video
A: audio
S: set
L: language
T: time-series
Ta: tables
F: force sensor
P: proprioception sensor

common architecture

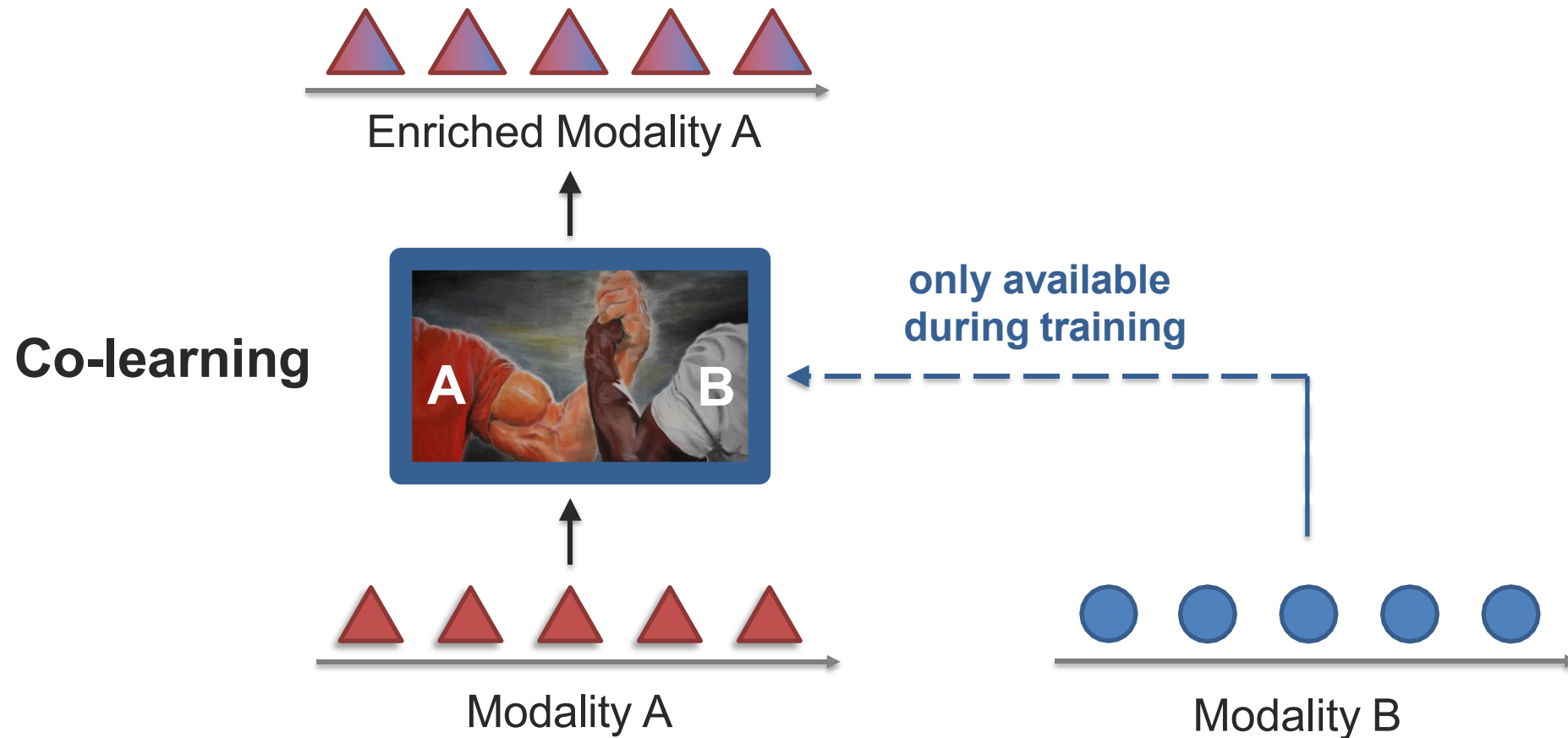
parameter sharing

Open challenges:

- Low-resource: little downstream data, lack of paired data, robustness (next section)
- Beyond language and vision
- Settings where SOTA unimodal encoders are not deep learning e.g., tabular data
- Complexity in data, modeling, and training
- Interpretability (next section)

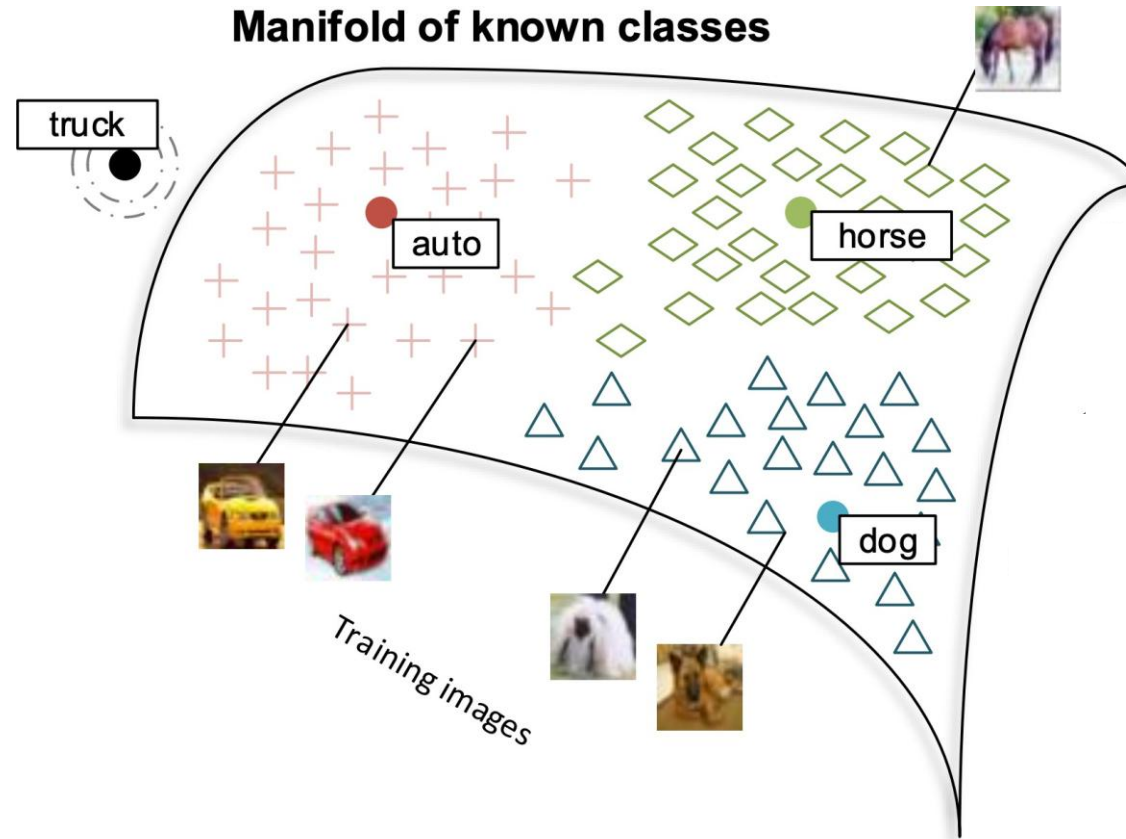
Sub-Challenge 5b: Co-learning

Definition: Transferring information from secondary to primary modality by sharing representation spaces between both modalities.

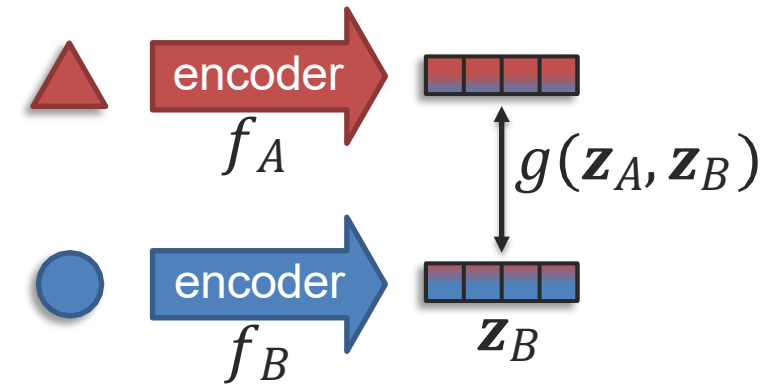


Co-learning via Representation

Representation coordination: word embedding space for zero-shot visual classification

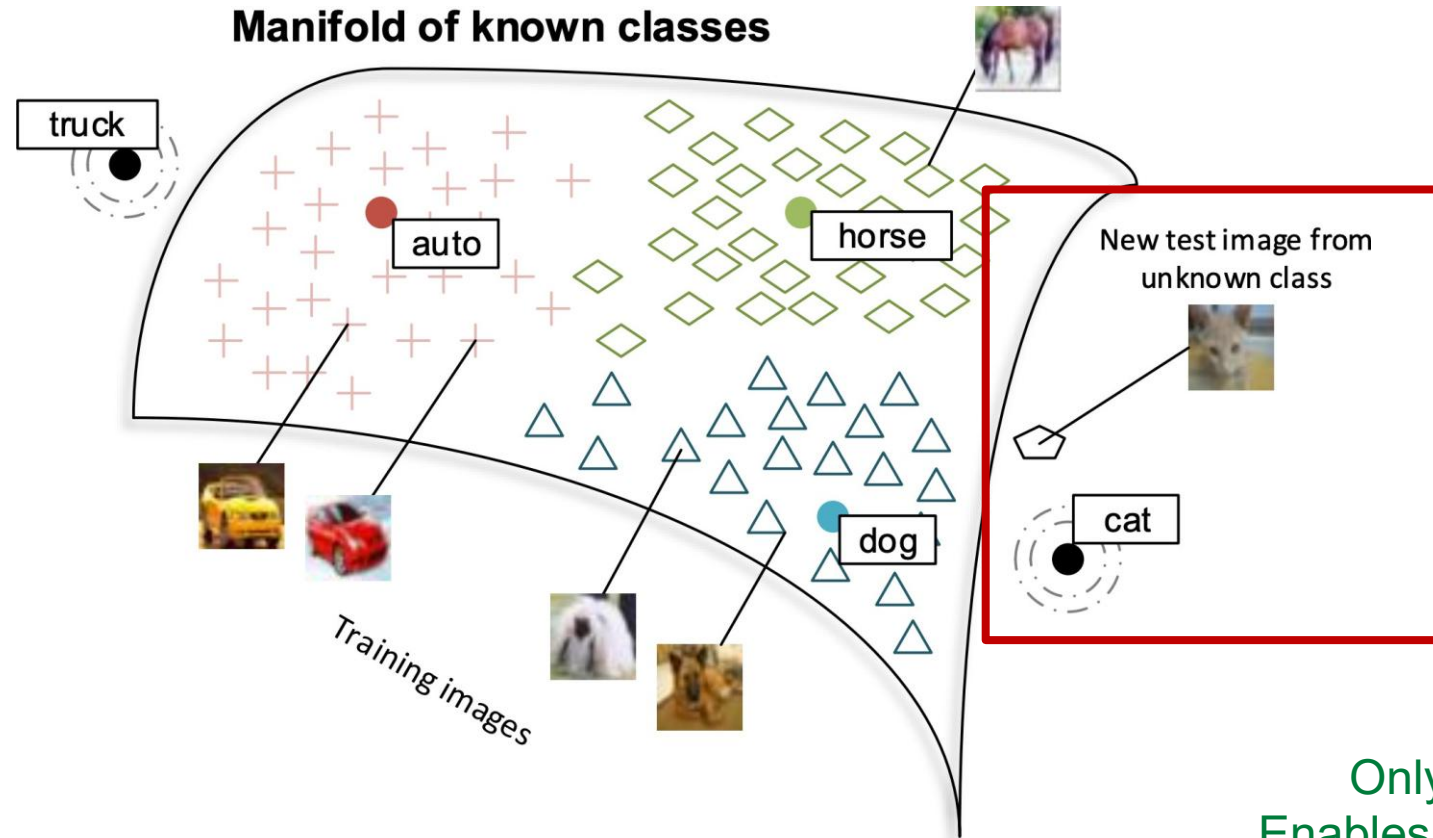


Recall representation coordination!

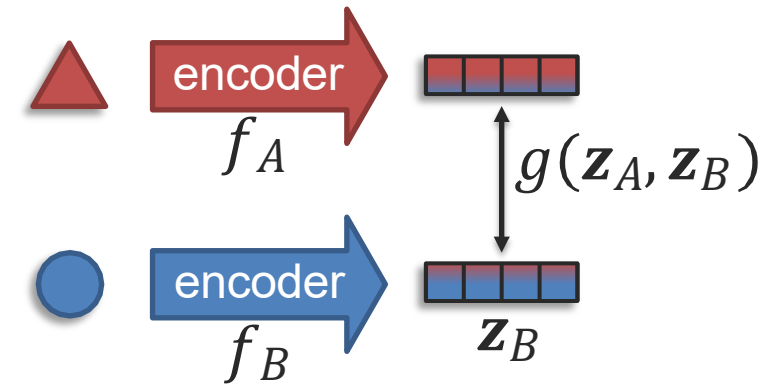


Co-learning via Representation

Representation coordination: word embedding space for zero-shot visual classification



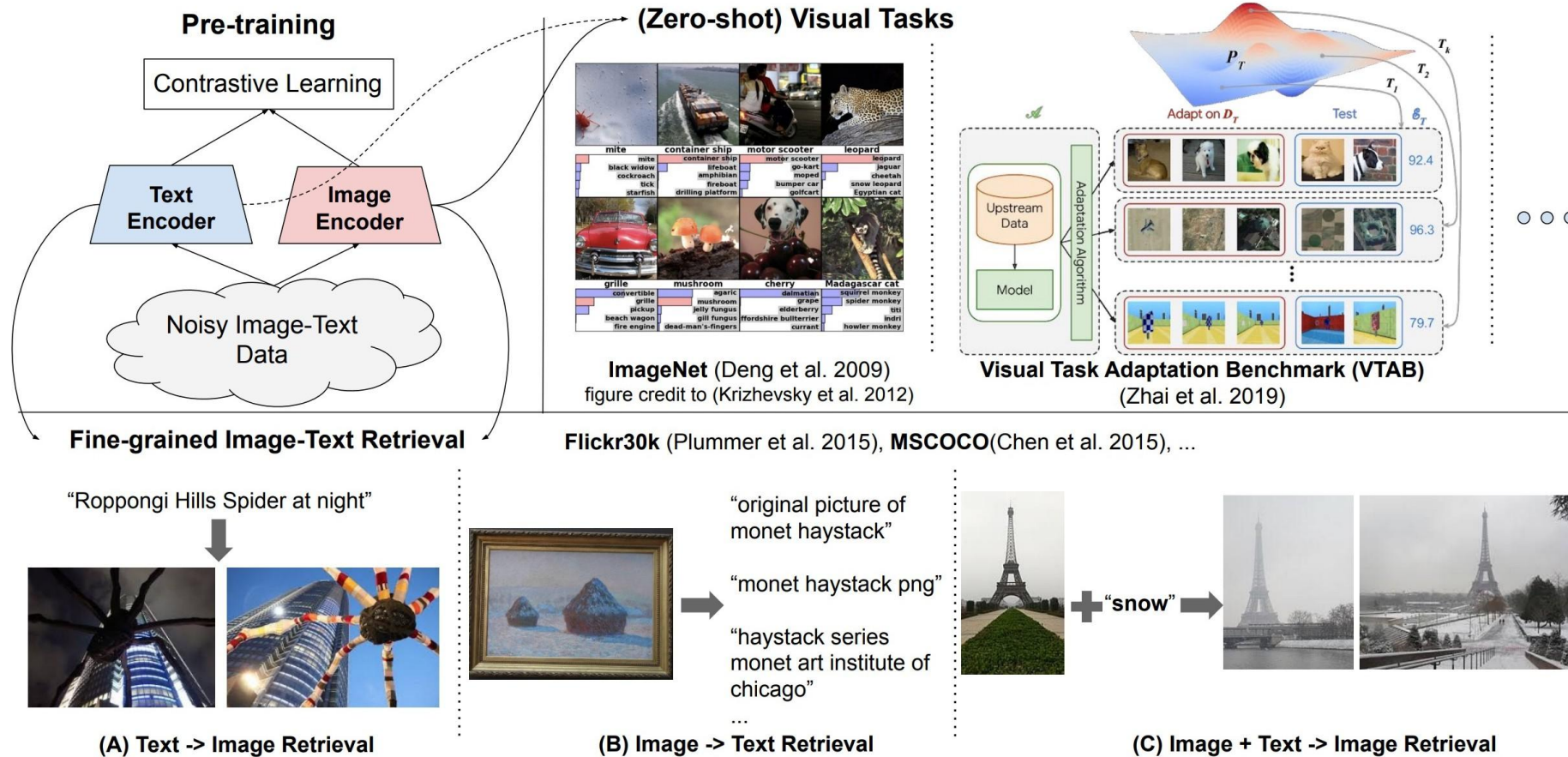
Recall representation coordination!



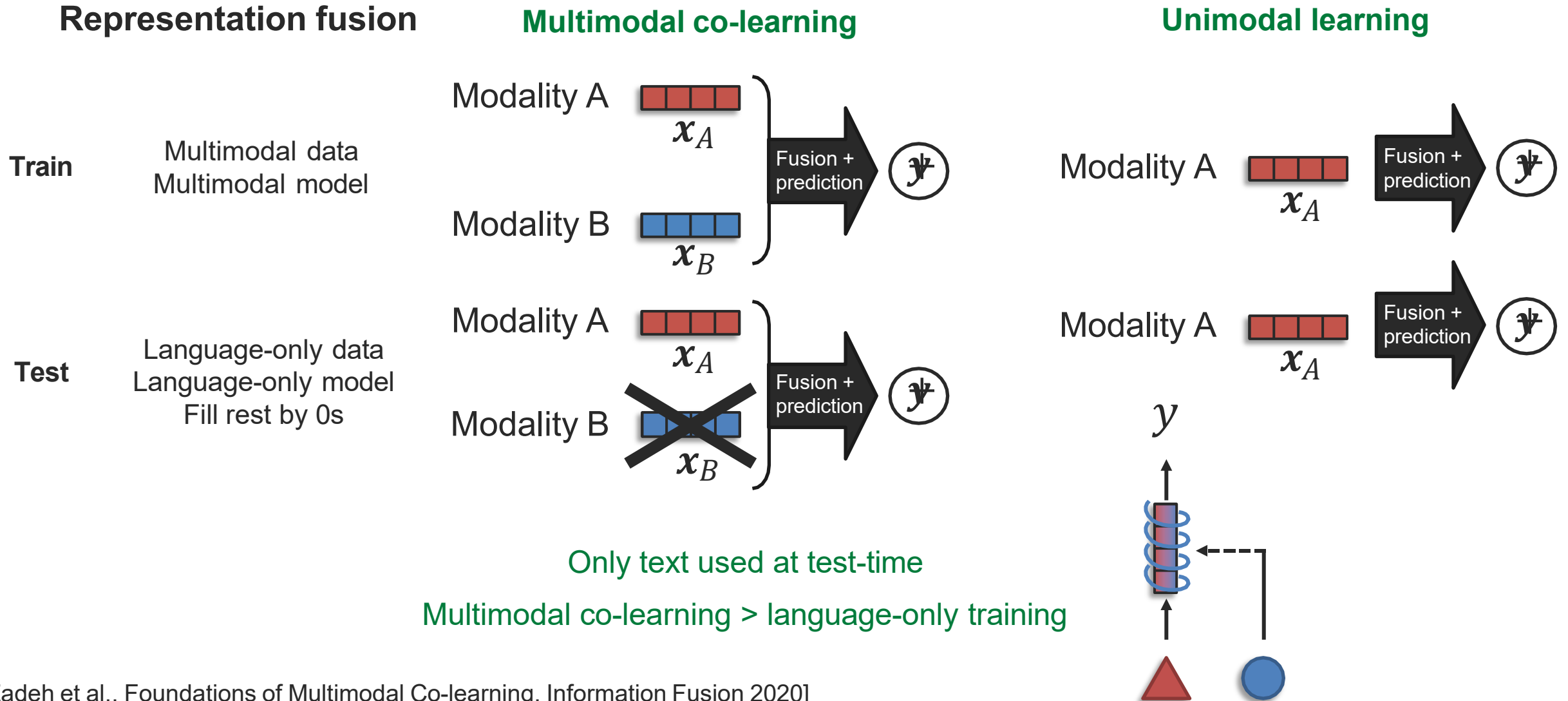
Only images used at test-time
Enables zero-shot image classification

Co-learning via Representation

Representation coordination at scale

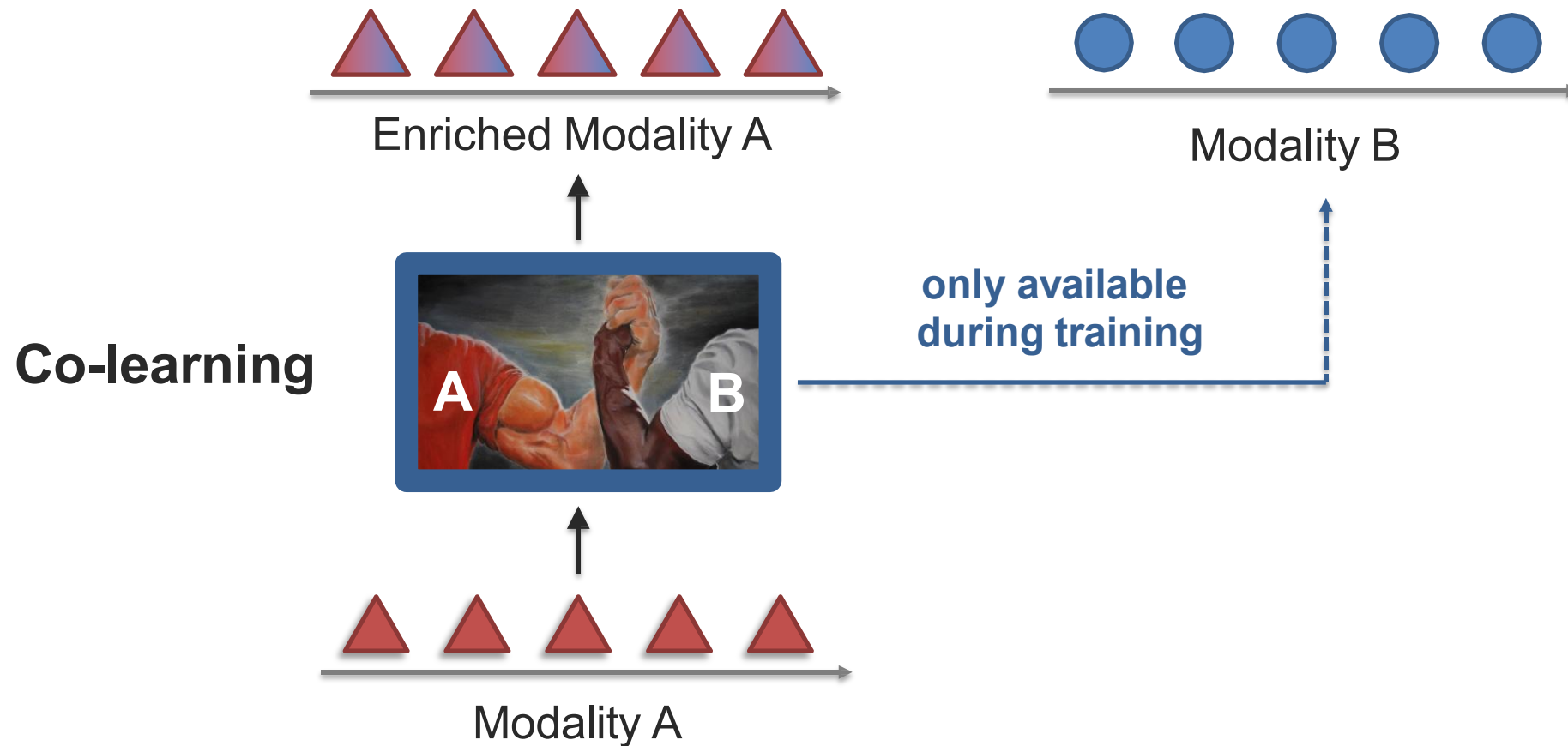


Co-learning via Representation



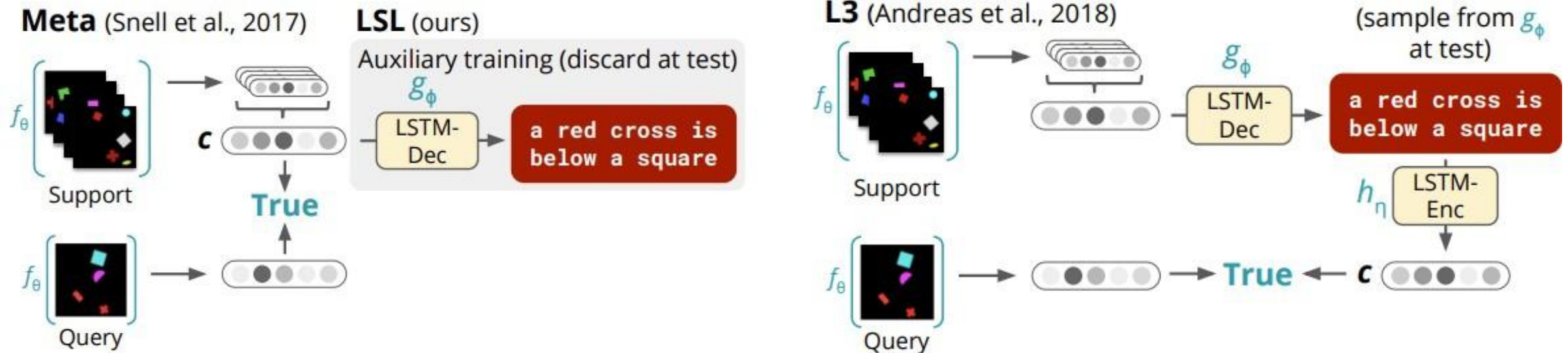
Co-learning via Generation

Definition: Transferring information from secondary to primary modality by using the secondary modality as a generation target.



Co-learning via Generation

Image to text generation



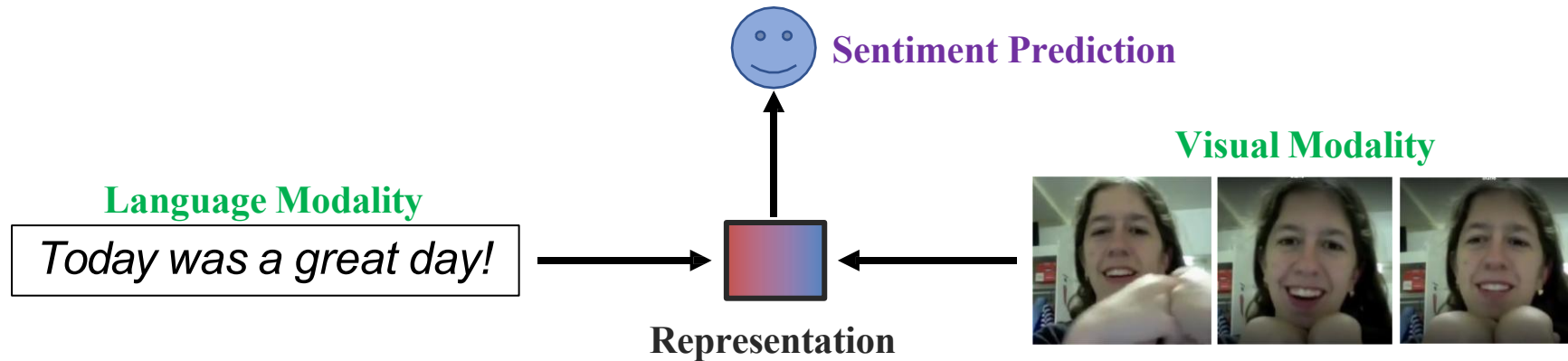
[Mu et al., 2019. Shaping Visual Representations with Language for Few-Shot Classification]

[Andreas et al. 2017, Learning with Latent Language]

[Sharma et al. 2021. Skill Induction and Planning with Latent. Language]

Co-learning via Generation

Bimodal translations

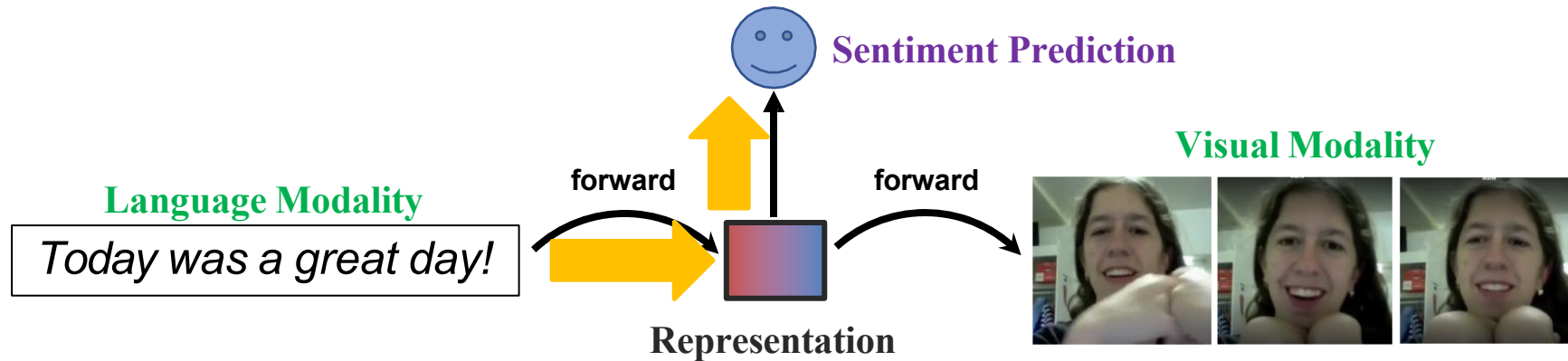


Both modalities required at test time!
Sensitive to noisy/missing visual modality.

We want to leverage information from visual modality
while being robust to it during test-time.

Co-learning via Generation

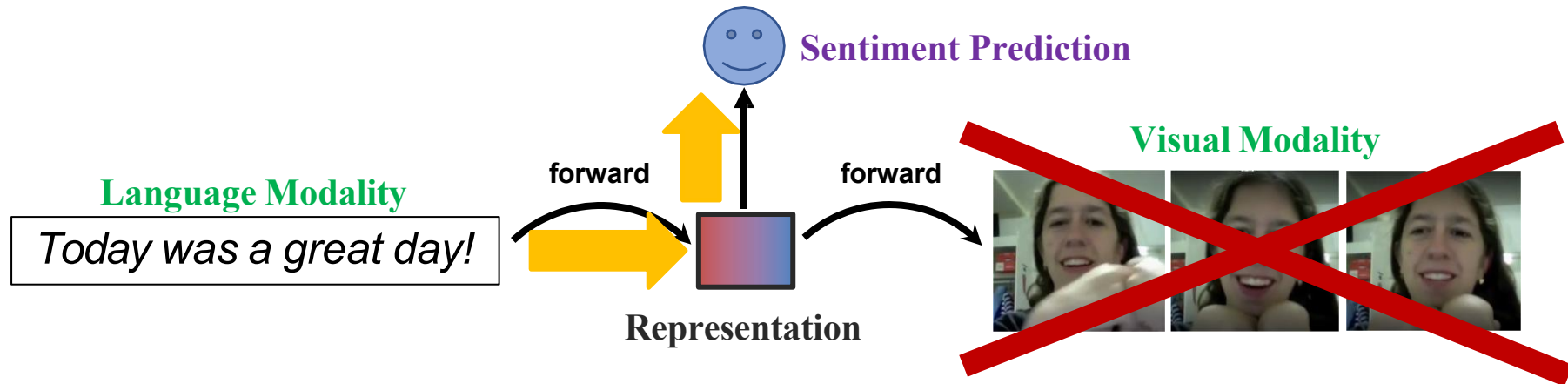
Bimodal translations



Cross-modal translation during training
Only language modality required at test time!

Co-learning via Generation

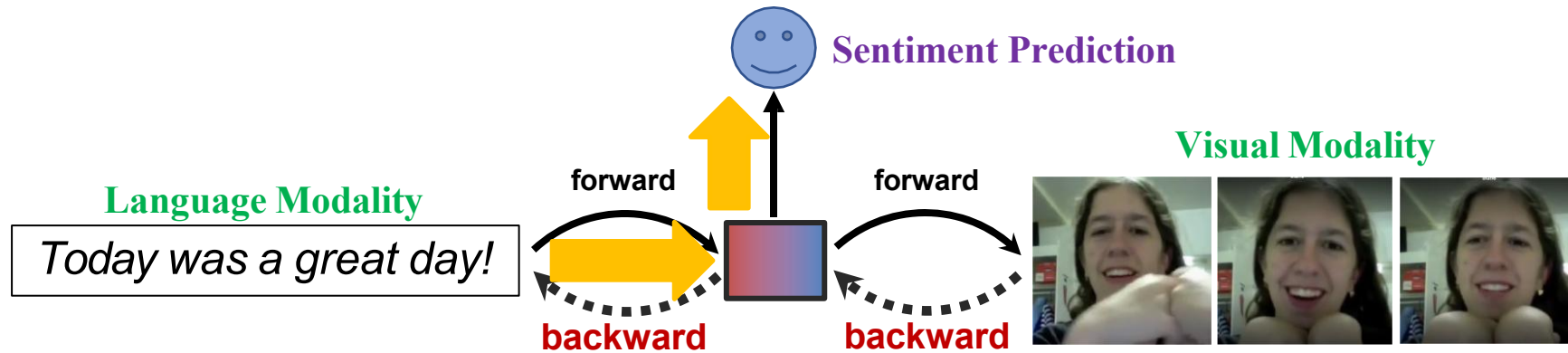
Bimodal translations



Problem: how do you ensure that both modalities are being used?

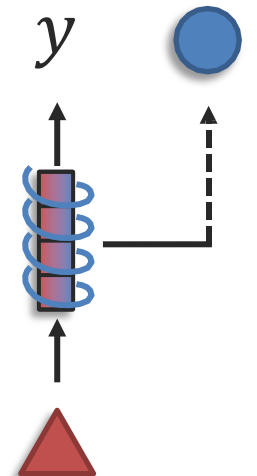
Co-learning via Generation

Bimodal cyclic translations



Solution: cyclic translations from visual back to language

Cross-modal translation during training
Only language modality required at test time!



Co-learning via Generation

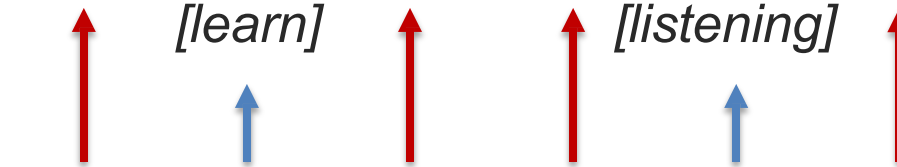
Predicting images from corresponding language

Voken (visual token) classification

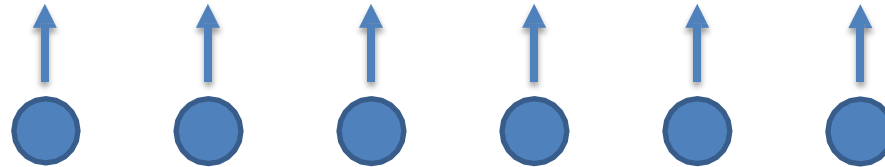


Masked language modeling

[learn] [listening]



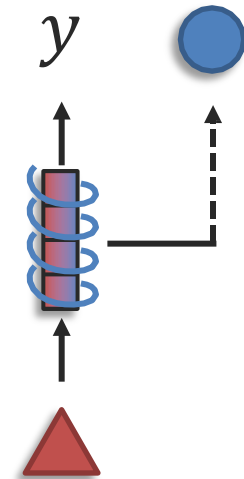
BERT language model



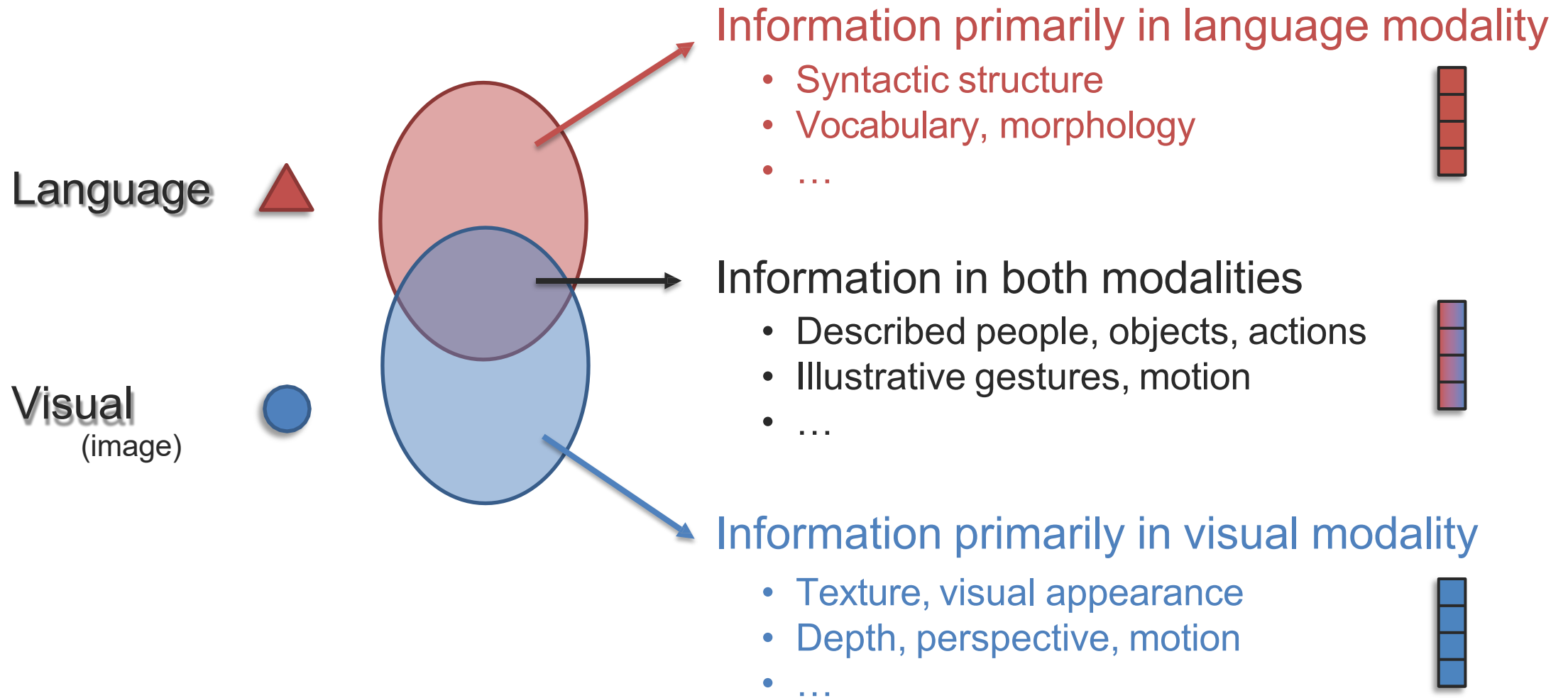
Humans [mask] language by [mask] speaking

Only text used at test-time

Multimodal co-learning > language-only training



Co-learning via Generation



Co-learning may not always work...

Open
challenges

Vision-language pretrained models on lexical grounding

Sentence-level semantic tasks

Encoder	SRL	Coref.	SPR	Rel.
BERT _{base}	90.10 $\pm$ 0.20	95.90 $\pm$ 0.00	83.70 $\pm$ 0.00	76.25 $\pm$ 0.05
VideoBERT _{text}	84.33 $\pm$ 0.05	92.47 $\pm$ 0.05	78.23 $\pm$ 0.05	65.83 $\pm$ 0.21
VideoBERT _{VL}	84.73 $\pm$ 0.05	92.82 $\pm$ 0.05	78.80 $\pm$ 0.00	66.37 $\pm$ 0.80
VisualBERT _{text}	89.00 $\pm$ 0.00	94.87 $\pm$ 0.05	82.27 $\pm$ 0.05	74.37 $\pm$ 0.19
VisualBERT _{VL}	89.57 $\pm$ 0.21	95.13 $\pm$ 0.05	82.17 $\pm$ 0.09	74.83 $\pm$ 0.05

Not much improvements with visual co-learning

Semantic Role Labeling “The **carrots** are then pureed in the food processor”
Entity Coreference “After the **apples** are chopped, put **them** in the bowl”

Co-learning may not always work...

Open
challenges

Vision-language pretrained models on seemingly multimodal tasks

Physical commonsense QA

Encoder	Linear	MLP	Trans.
BERT _{base}	55.43 $\pm$ 0.31	57.98 $\pm$ 0.16	60.12 $\pm$ 1.43
VideoBERT _{text}	57.87 $\pm$ 0.64	58.97 $\pm$ 0.44	62.35 $\pm$ 1.23
VideoBERT _{VL}	58.51 $\pm$ 0.20	58.56 $\pm$ 0.27	63.66 $\pm$ 1.31
VisualBERT _{text}	54.81 $\pm$ 0.19	56.81 $\pm$ 0.24	58.63 $\pm$ 0.79
VisualBERT _{VL}	55.83 $\pm$ 0.27	59.10 $\pm$ 0.11	61.66 $\pm$ 1.08

Marginal improvements with visual co-learning

“Remove gloss from furniture.”

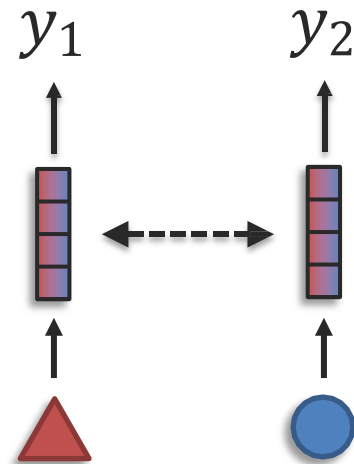


“Rub furniture with steel wool/cotton ball”

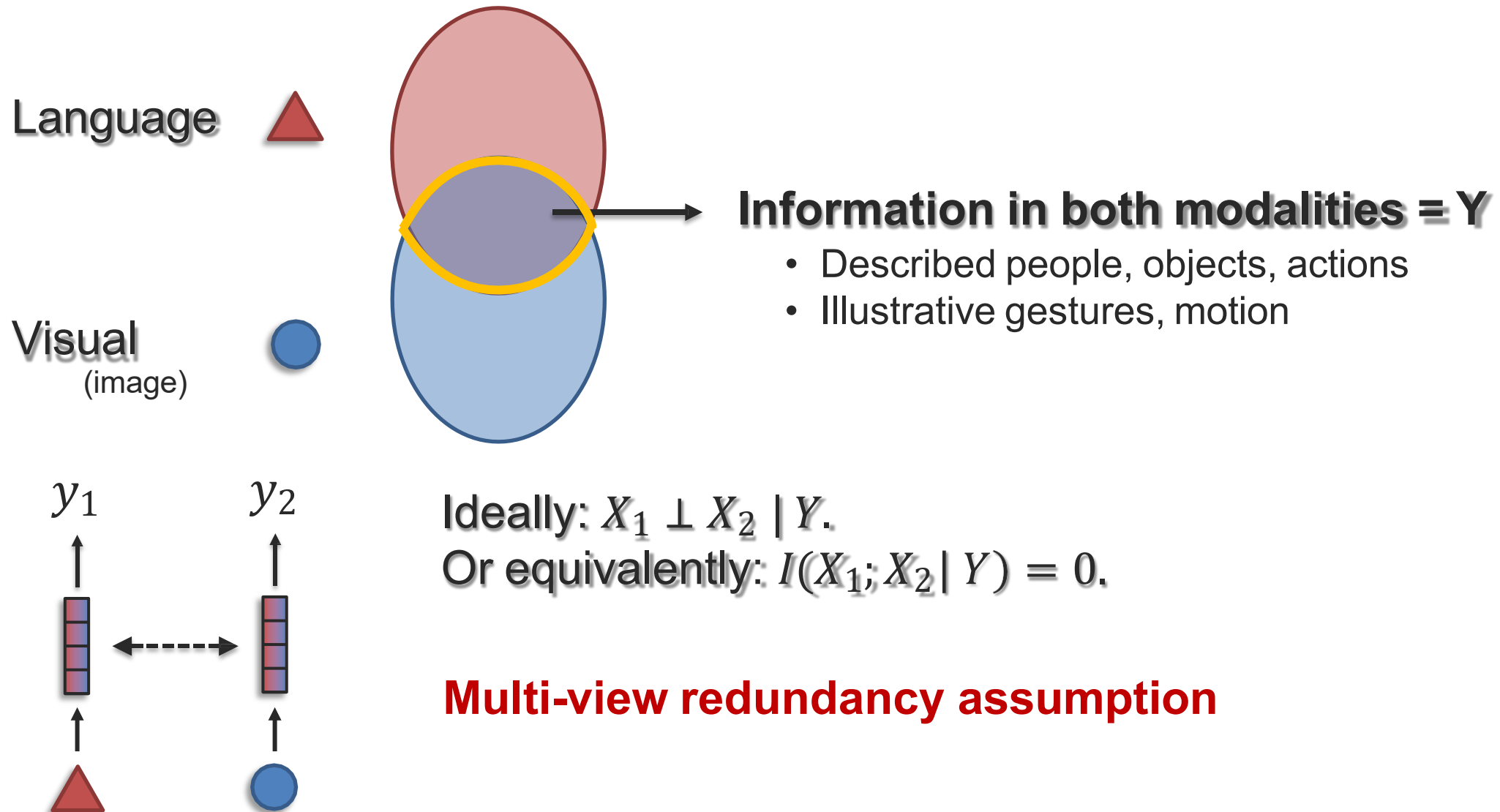
Sub-challenge 5c: Model Induction

Definition: Keeping individual unimodal models separate but inducing common behavior across separate models.

Model Induction



Sub-challenge 5c: Model Induction

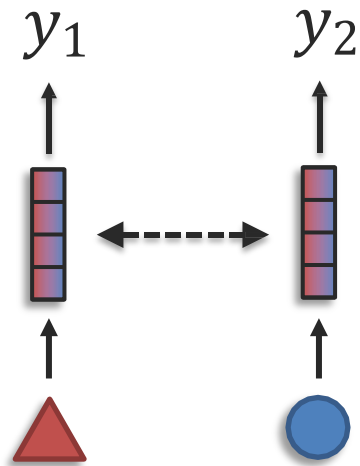


Co-training

Setup

Ideally: $X_1 \perp X_2 \mid Y$.

Or equivalently: $I(X_1; X_2 \mid Y) = 0$.



Multi-view redundancy assumption

1. X_1 = text on the web page.
2. X_2 = text on hyperlinks pointing into the web page.
3. Y = category of web page: academic, sports, news, music etc.

Sufficiency assumption

$X_1 \rightarrow Y$ is learnable given enough data

$X_2 \rightarrow Y$ is learnable given enough data

Self-training

Warmup: a single view – Self-training



Assume:

1. Labeled data $\{X_1^L, Y\}$.
2. Unlabeled data $\{X_1^U\}$.

Train:

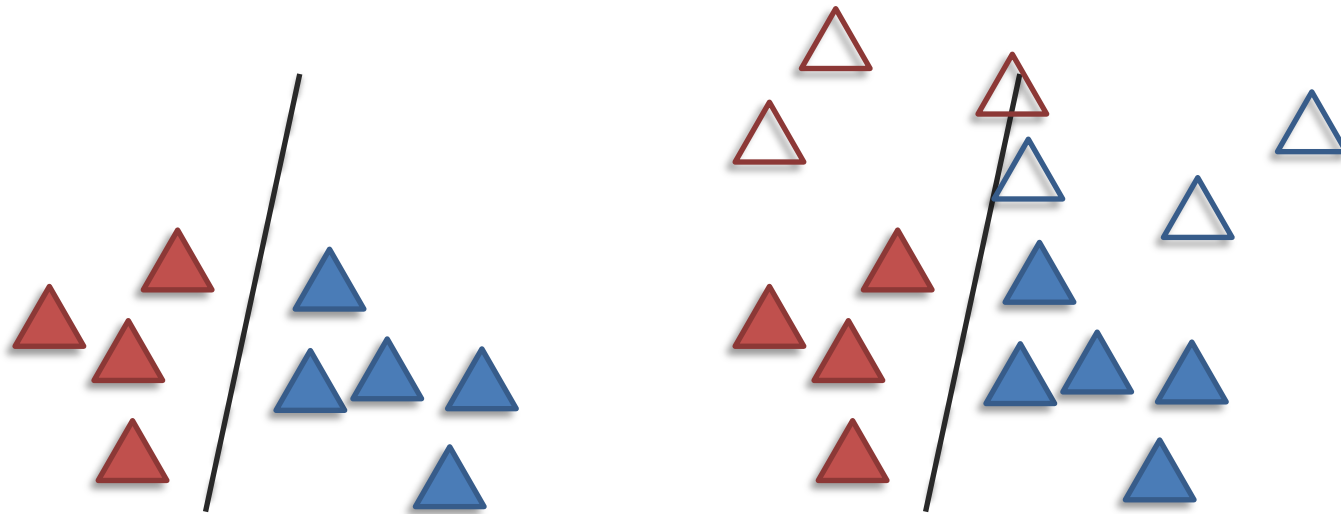
1. Train classifier f_1 on $\{X_1^L, Y\}$.
2. Use classifier f_1 to label the most confident examples in $\{X_1^U\}$ and add it to the labeled set $\{X_1^U, Y = f_1(X_1^U)\}$.
3. Go to 1, and repeat until there are no more unlabeled samples.

Test:

1. For a new unlabeled sample $\{X_1\}$, output $f_1(X_1)$.

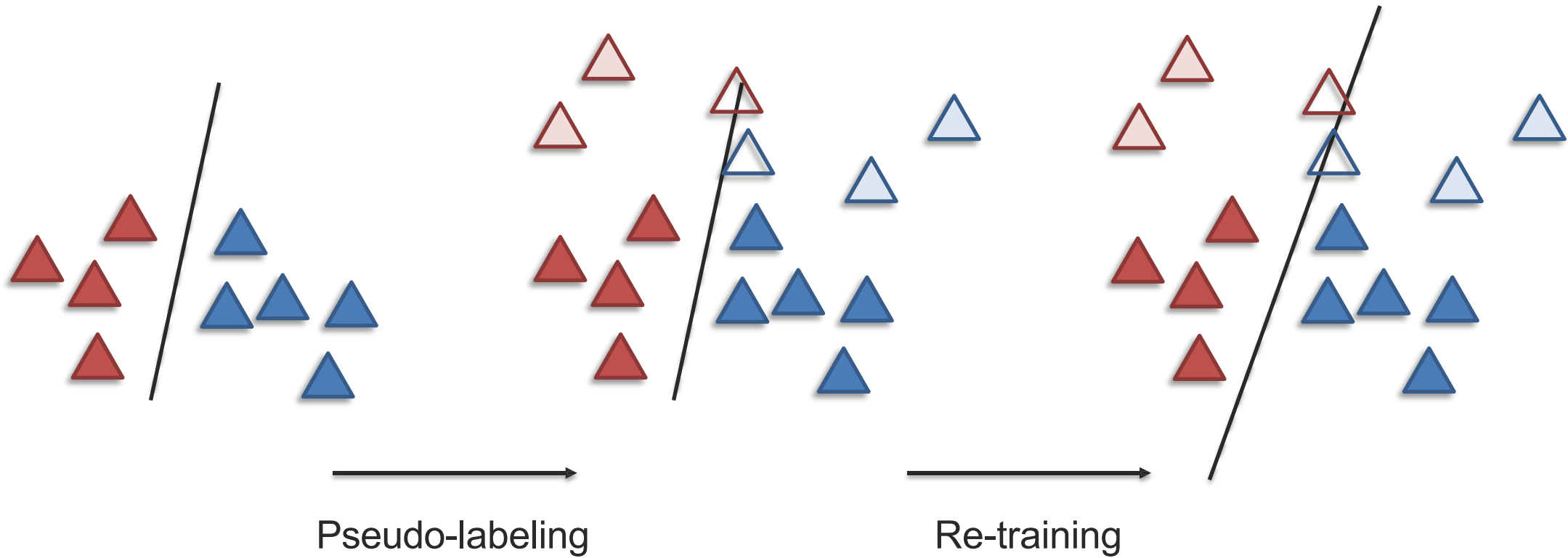
Self-training

Warmup: a single view – Self-training



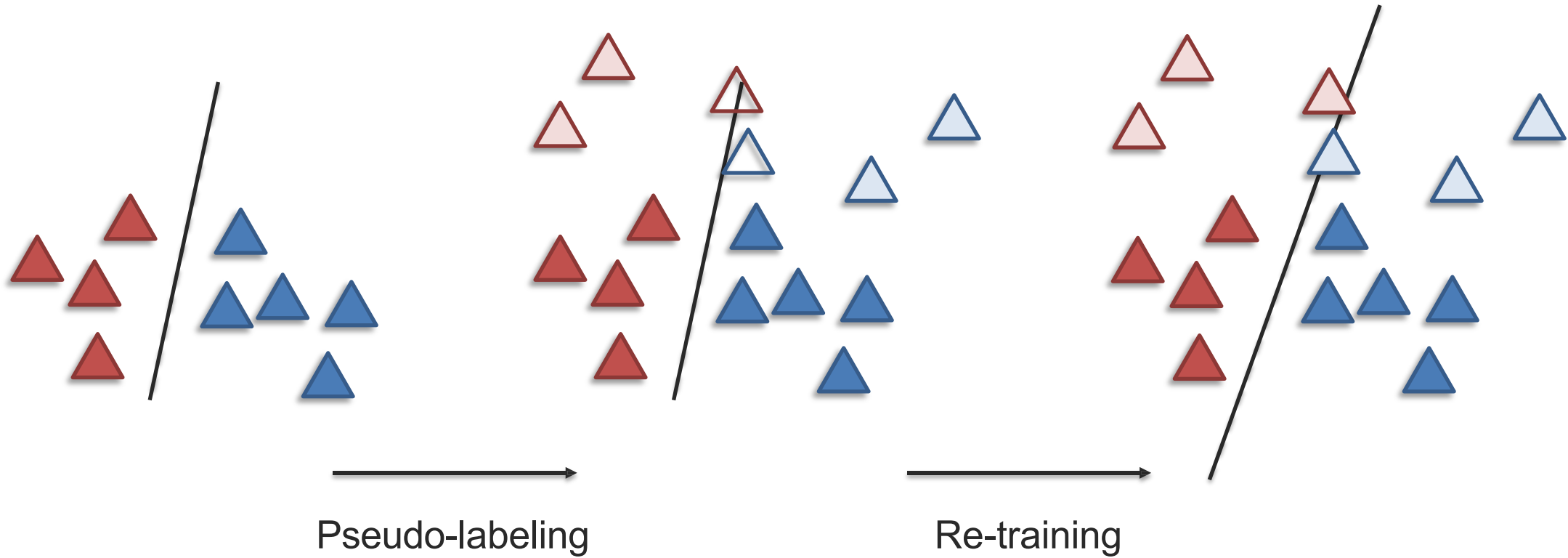
Self-training

Warmup: a single view – Self-training



Self-training

Warmup: a single view – Self-training

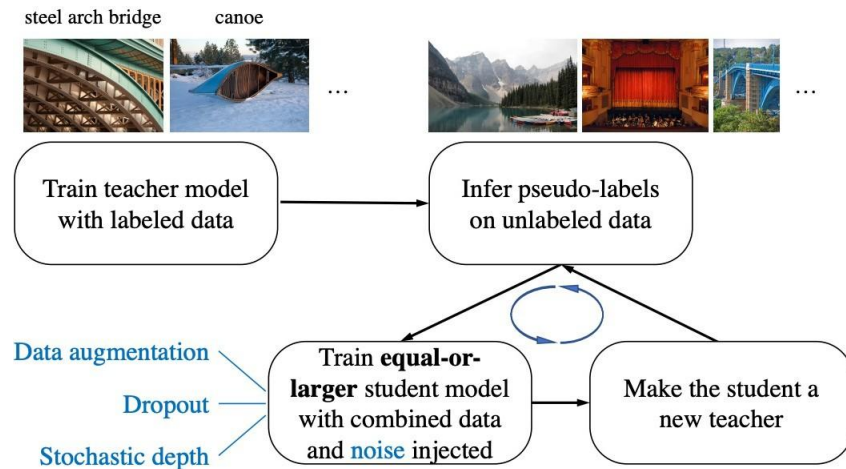


Self-training

Key-words: semi-supervised learning, label propagation, domain adaptation/shift

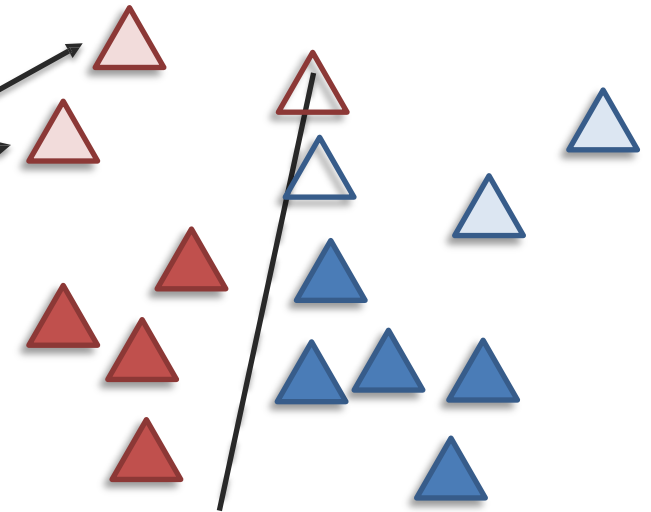
Critical:

1. Can't label all unlabeled data in one step, or you recover original classifier just trained on labeled data.
2. Sequence of pseudo-labeling is important to gradually shift classification boundary.
3. Input consistency regularization: shape of data space is important – implicit assumption that similar datapoints have similar labels (i.e., label consistency)



Input consistency:

- Data augmentation
- Adding noise



Co-training

From self-training to co-training

Ingredients:

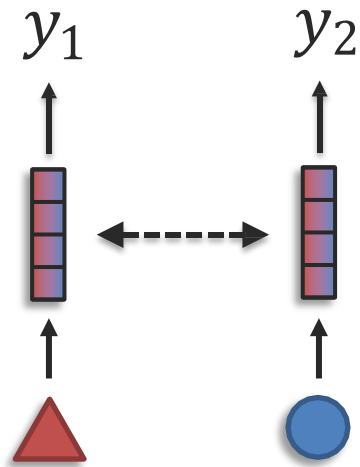
- Two views on the data: x_1 and x_2
- Two classifiers: $x_1 \rightarrow y$ and $x_2 \rightarrow y$
- A bit of labeled data (x_1, x_2, y) ; lots of unlabeled data (x_1, x_2)

Assumptions:

1. Either view is sufficient to predict the label alone, with enough data
2. Views should be as independent as possible: examples where f_1 has high confidence but not f_2 and vice-versa.

Co-training

Algorithm



Assume:

1. **Small** amount of labeled data $\{X_1^L, X_2^L, Y\}$.
2. **Lots** of unlabeled data $\{X_1^U, X_2^U\}$.

Train:

1. Train classifier f_1 on $\{X_1^L, Y\}$ and f_2 on $\{X_2^L, Y\}$.
2. Use classifier f_1 to label the most confident examples in $\{X_1^U\}$ and add it to the labeled set to train f_2 $\{X_2^U, Y = f_1(X_1^U)\}$.
3. Use classifier f_2 to label the most confident examples in $\{X_2^U\}$ and add it to the labeled set to train f_1 $\{X_1^U, Y = f_2(X_2^U)\}$.
4. Go to 1, and repeat until there are no more unlabeled samples.

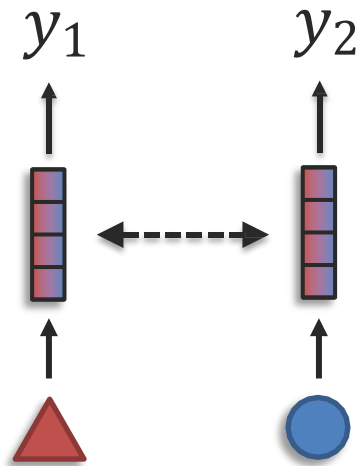
Test:

1. For a new unlabeled sample $\{X_1, X_2\}$, ensemble $f_1(X_1)$ and $f_2(X_2)$.

Co-training

Co-training

1. X_1 = text on the web page, X_2 = text on hyperlinks pointing into the web page.
2. Y = category of web page: academic, sports, news, music etc.



Louis-Philippe Morency

Leonardo Associate Professor of Computer Science,
[Language Technology Institute](#),
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I am tenure-track Faculty at CMU Language Technology Institute where I lead the Multimodal Communication and Machine Learning Laboratory ([MultiComp Lab](#)). I was previously Research Faculty at USC Computer Science Department. I received my Ph.D. in Computer Science from MIT Computer Science and Artificial Intelligence Laboratory.

My research focuses on building the computational foundations to enable computers with the abilities to analyze, recognize and predict subtle human communicative behaviors during social interactions. Central to this research effort is the technical challenge of multimodal machine learning: mathematical foundation to study heterogeneous multimodal data and the contingency often found between modalities. This multi-disciplinary research topic overlaps the fields of multimodal interaction, social psychology, computer vision, machine learning and artificial intelligence, and has many applications in areas as diverse as medicine, robotics and education.

Labeled, learn that ' $X_1(\text{LP}) = \text{CMU} \rightarrow \text{academic}$ ' and ' $X_2(\text{Paul} \rightarrow \text{LP}) = \text{advised by} \rightarrow \text{academic}$ '



Paul Pu Liang

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Machine Learning Department and Language Technologies Institute, School of Computer Science, Carnegie Mellon University

[CV] [@pliang279](#) [@pliang279](#) [@lpwinniethepu](#)

I am a fourth-year Ph.D. student in the Machine Learning Department at Carnegie Mellon University, advised by Louis-Philippe Morency and Ruslan Salakhutdinov. I also collaborate closely with Manuel Blum, Lenore Blum, and Daniel Rubin at Berkeley and Stanford. My research lies in the foundations of multimodal machine learning with applications in socially intelligent AI, understanding human and machine intelligence, natural language processing, healthcare, and education. As steps towards this goal, I work on:



Unlabeled, label using ' $f_1: X_1(\text{Paul}) = \text{CMU} \rightarrow \text{academic}$ ' and learn that ' $X_2(\text{MLD} \rightarrow \text{Paul}) = \text{PhD program} \rightarrow \text{academic}$ '

Another student $\rightarrow$ **Unlabeled**, label using ' $f_2: X_2(\text{Berkeley CS} \rightarrow \text{student}) = \text{'PhD program} \rightarrow \text{academic}'$ ' and learn that ' $X_1(\text{student}) = \text{robotics} \rightarrow \text{academic}$ '

Co-training

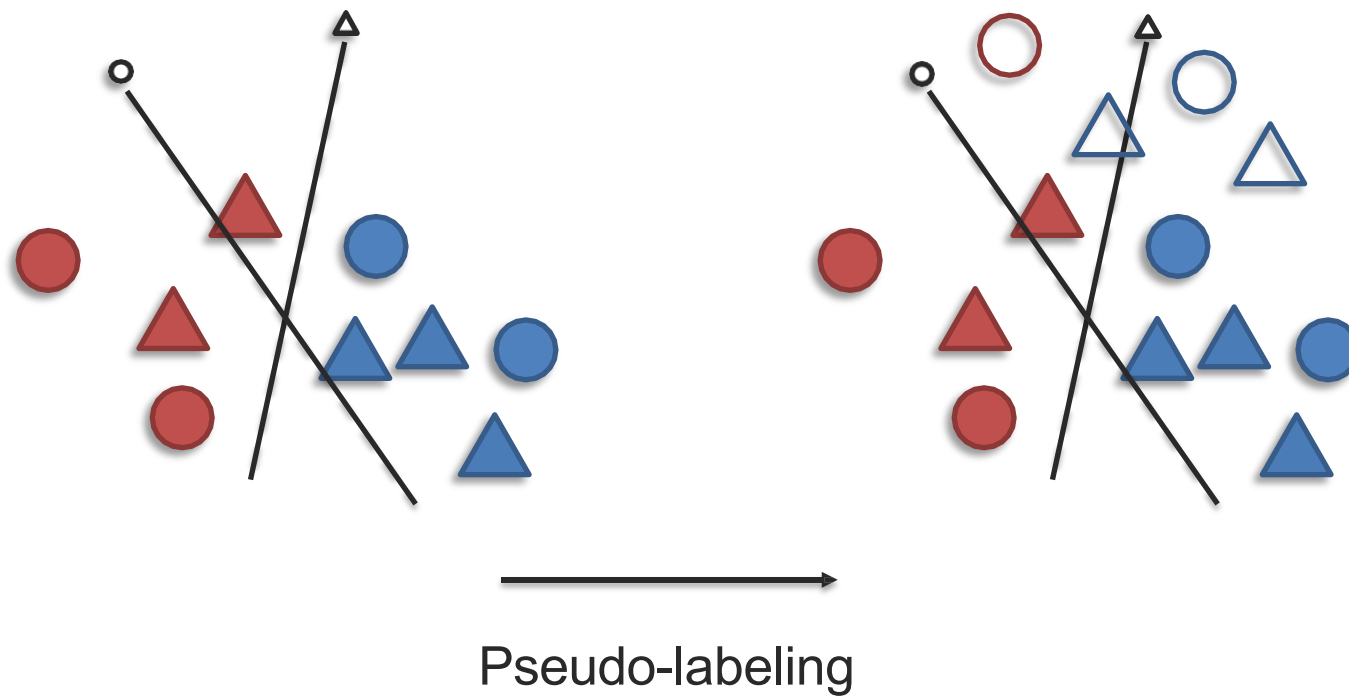
From self-training to co-training

Assumptions:

1. Either view is sufficient to predict the label alone.
2. Views should be as independent as possible: examples where f_1 has high confidence but not f_2 and vice-versa.

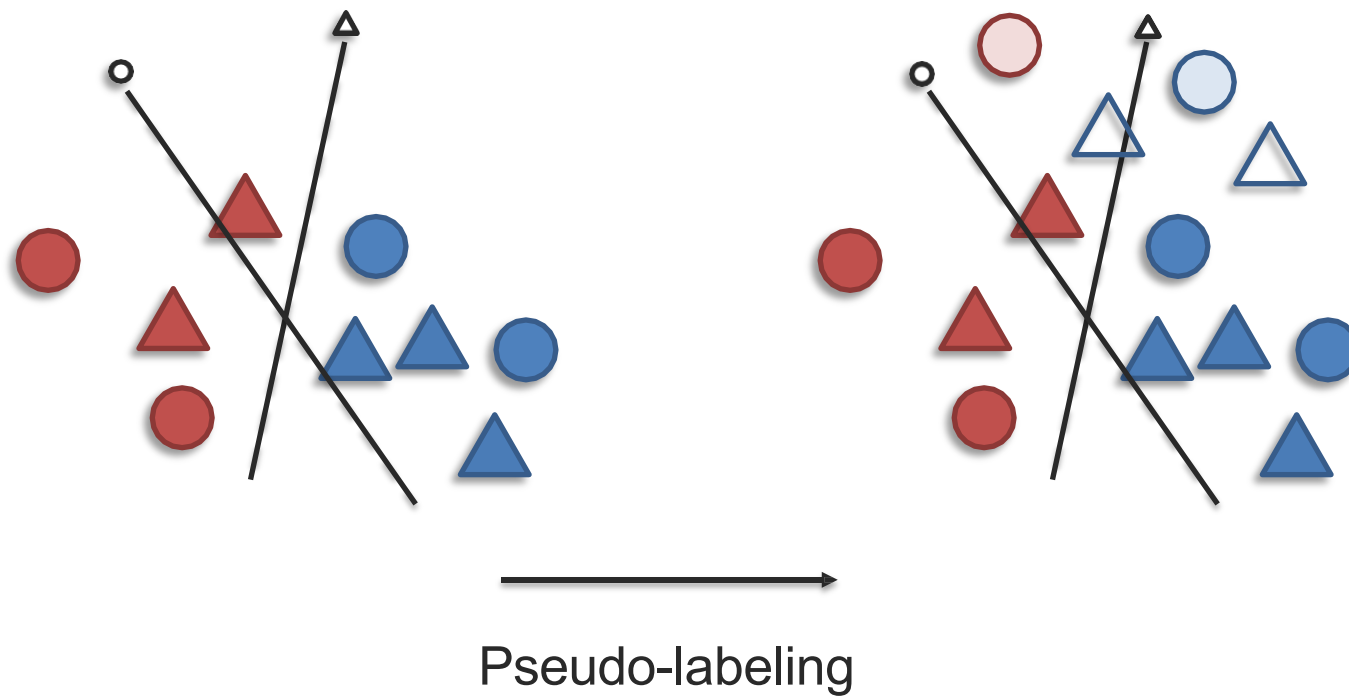
Co-training

From self-training to co-training



Co-training

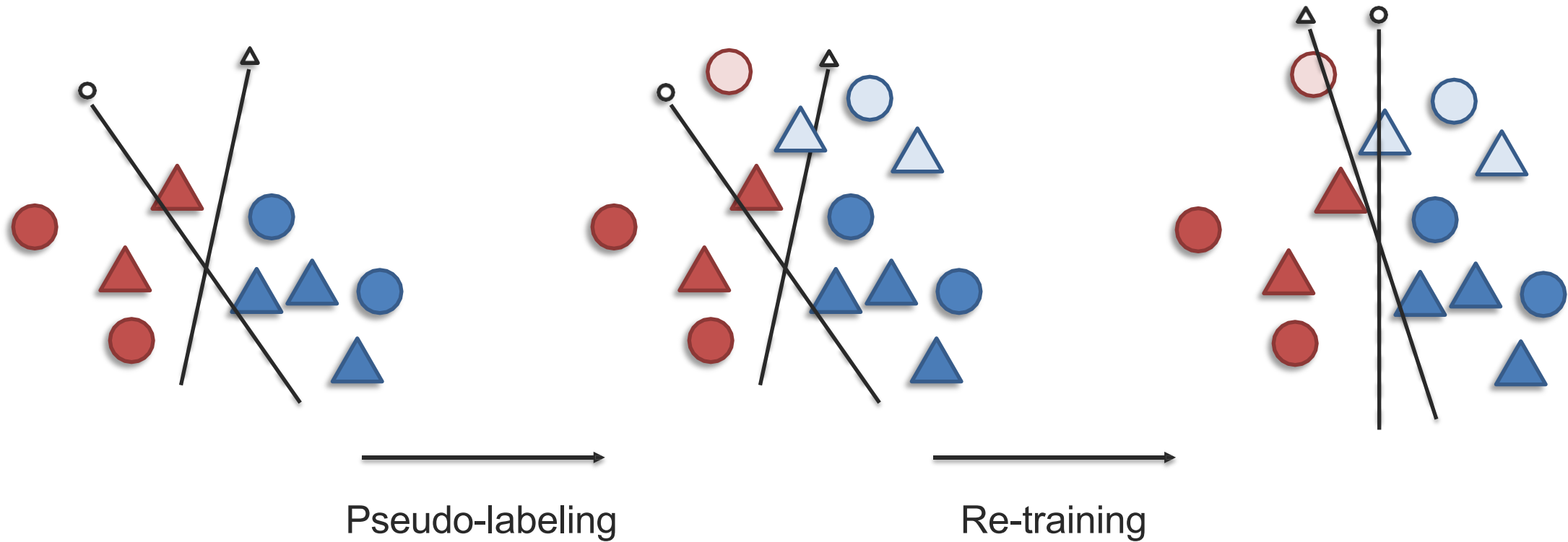
From self-training to co-training



Co-training

From self-training to co-training

Key idea: functions on both views must be compatible and agree



Co-training

From self-training to co-training

Key idea: functions on both views must be compatible and agree

Intuitions:

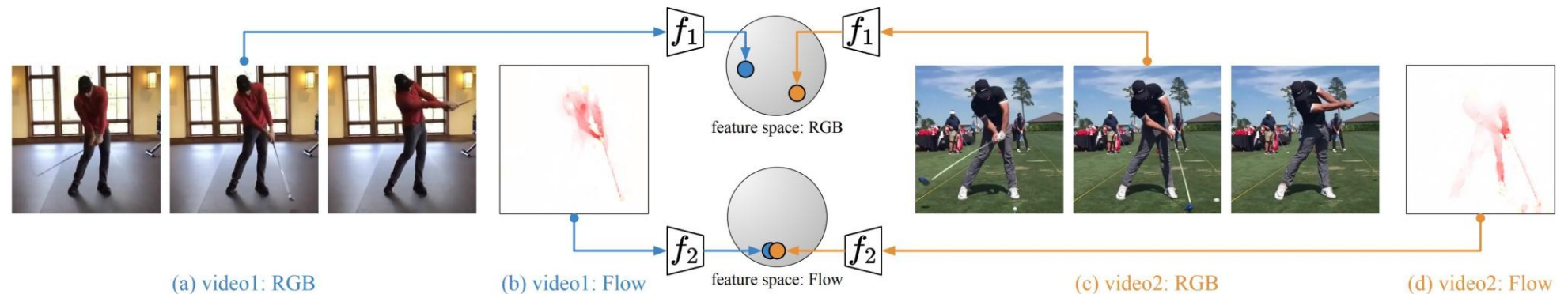
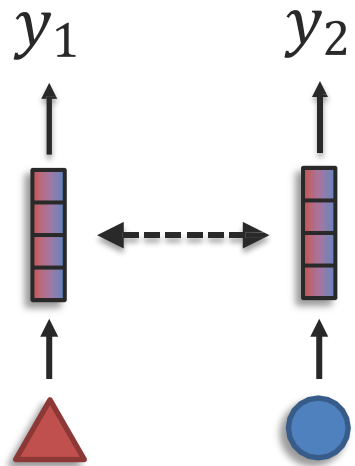
1. Either view is sufficient to predict the label alone.
2. Views should be as independent as possible: examples where f_1 has high confidence but not f_2 and vice-versa.
3. Input consistency regularization: shape of data space is important – implicit assumption that similar datapoints have similar labels (i.e., label consistency).
 - In co-training, data from another view help us to supplement the label space!
 - Both views must agree = input consistency which enables cross-view pseudo-labeling.
4. Eventually, will converge on 2 classifiers that agree and each separate both views.

Co-training

Recent applications of co-training

Self-supervised learning with positive and negative samples

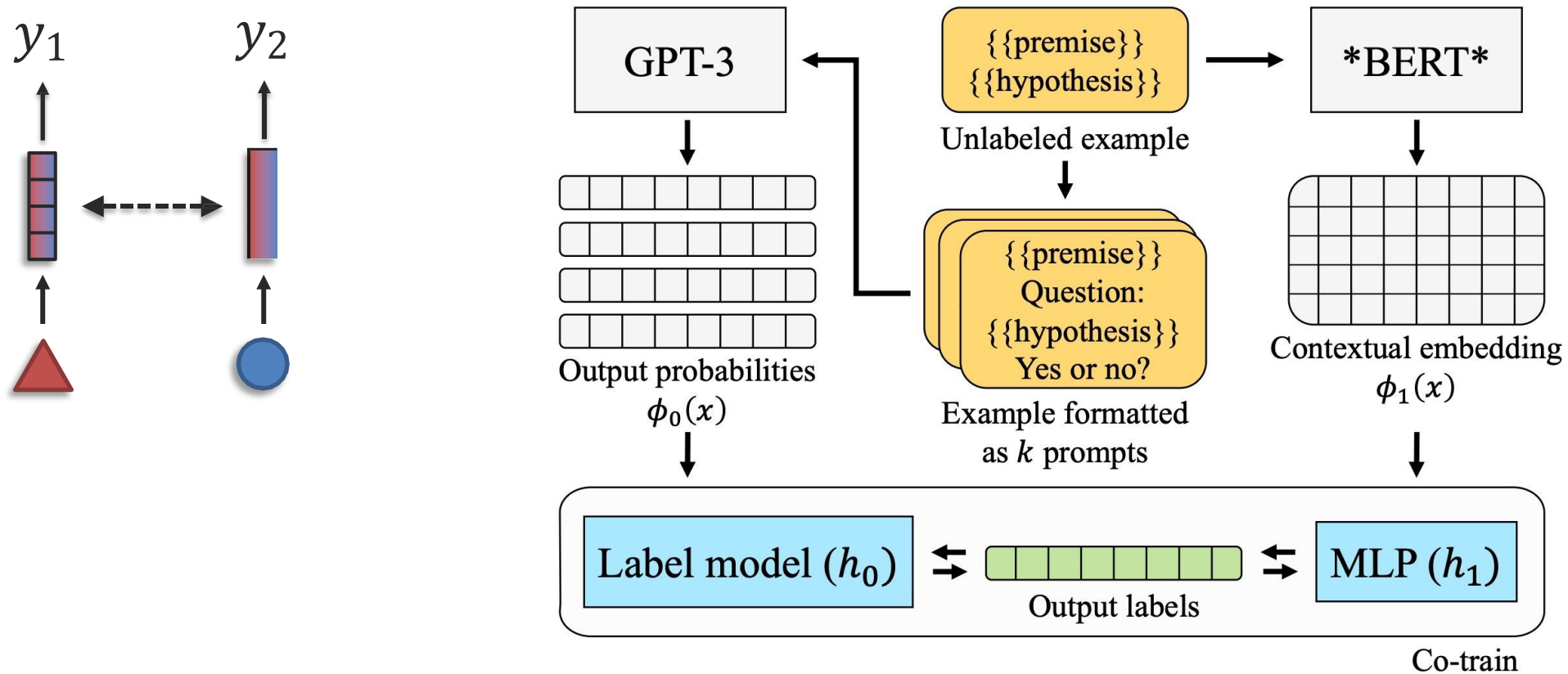
- Positive samples hard to discover in RGB space can be easily found in flow space, and vice-versa (e.g., RGB sensitive to background differences but not flow).
- Can use co-training between 2 RGB and flow contrastive learning modules.



Co-training

Recent applications of co-training

Language-model prompting



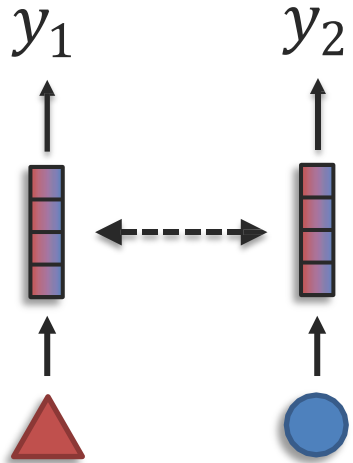
Co-Regularization

Co-regularization

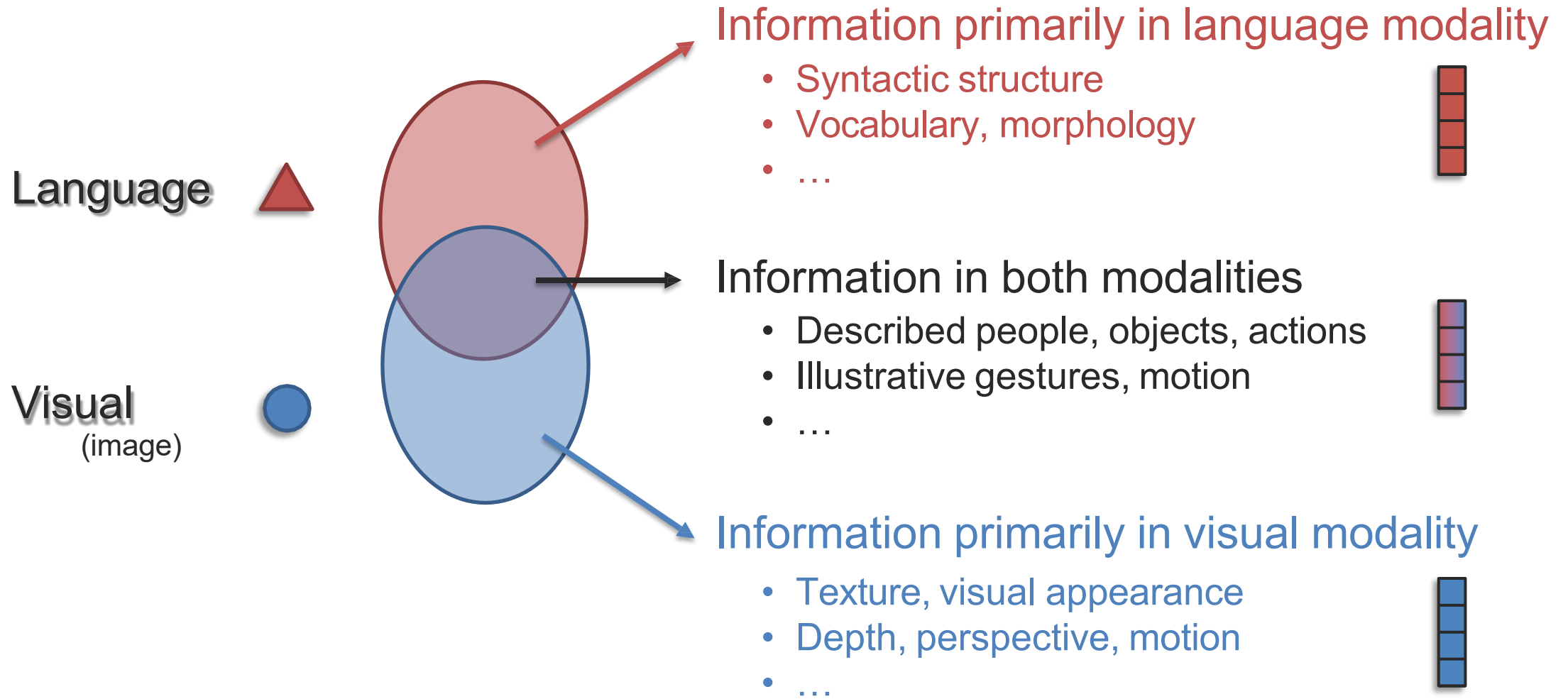
Add a loss term to ensure both model predictions are similar:

$$L = (f_1(X_1) - f_2(X_2))^2$$

Recall representation coordination.



Sub-challenge 5c: Model Induction

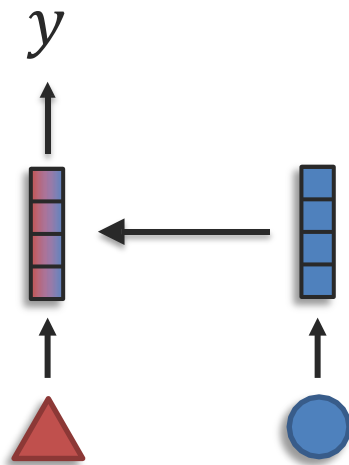


Summary: Transference

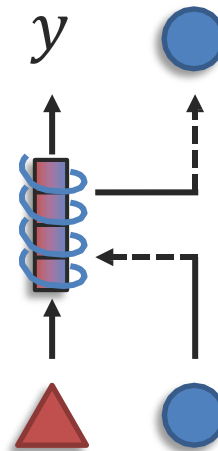
Definition: Transfer knowledge between modalities, usually to help the primary modality which may be noisy or with limited resources.

Sub-challenges:

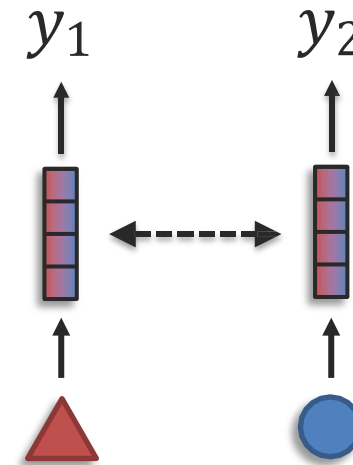
Transfer



Co-learning



Model Induction



More Transference



**Open
challenges**

Many more dimensions of transfer:

- **Multimodal {multitask, transfer, few-shot, meta} learning.**
- **Domain adaptation, domain shift, label shift.**
- **Core: representation, alignment, reasoning!**

Open challenges:

- Low-resource: little downstream data, lack of paired data, robustness (next section).
- Settings where SOTA unimodal encoders are not deep learning e.g., tabular data.
- Evaluating reasoning and robustness and large models.
- Limits of transfer beyond redundancy/joint information.
- Interpretability (next section).